GEODESIC STRING COUNTING INVARIANTS OF MANIFOLDS

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Abstract. We define rational valued geodesic-string counts for complete Riemann-Finsler manifolds using the Fuller index of the geodesic flow. We prove that this count is invariant under taut deformations of the metric, and prove a topological interpretation in the indivisible case via S^1-equivariant homology of the free loop space. As one application, for every sufficiently large C, we prove that for Finsler metrics sufficiently close to the flat metric on T^n, a class-β geodesic string of flat length $< C$ whose multiplicity is divisible by a prime p forces the existence of another such string. We also prove a fibration/product formula for the main count and use it to obtain Preissmann-type obstructions: under natural hypotheses on a fibration $Z \hookrightarrow X \to Y$, the total space admits neither a complete negatively curved Riemannian metric nor a forward complete Finsler metric with a unique nondegenerate geodesic string in the relevant fiber class. Finally, we show that sky catastrophes are not geodesible by a class of forward complete Riemann–Finsler metrics, including complete Riemannian metrics of nonpositive sectional curvature.

1. Introduction

Using counts of ***geodesic strings*** (equivalence classes of closed, constant speed geodesics up to reparametrization S^1 action), and the Fuller index [8], we will define certain rational number valued, deformation invariants for complete Riemann-Finsler manifolds. A basic case is a complete Riemannian manifold with non-positive sectional curvature, and in this case we get deformation invariants of the metric, provided the deformation is through non-positive sectional curvature metrics.

Studying connections of the Fuller index with Riemann-Finsler geometry was suggested by Fuller himself in the 1960's but was never much developed. An outline of Fuller's theory and basic definitions are given here in the Appendix.

Suppose for the moment X is compact. For a non-constant, free homotopy class β of a loop in X, we say that a metric g on X is β-***regular*** if all of its class β closed geodesics are non-degenerate the usual sense, or equivalently the closed orbits of the associated geodesic flow are dynamically non-degenerate. For a β-regular g, with finitely many class β geodesic strings, we have the following formula for the count:

$$F(g, \beta) = \sum_{o} \frac{(-1)^{\mathrm{morse}(o)}}{\mathrm{mult}(o)} \in \mathbb{Q}, \tag{1.1}$$

where:

- The sum is over class β g-geodesic strings o.
- $\mathrm{morse}(o)$ denotes the Morse index of o, (meaning the Morse-Bott index of the associated critical submanifold (diffeomorphic to S^1) of the loop space).
- $\mathrm{mult}(o)$ is the geometric multiplicity, equivalently the order of the corresponding isotropy subgroup of S^1, where S^1 is acting on the loop space by reparametrization.

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For a free homotopy class β of loops in X, let $L_\beta X$ denote the corresponding component of the free loop space. The following is a direct corollary of Theorem 1.24, and exemplifies the basic topological invariance of the count:

Corollary 1.2. *Let X, g, β be as above and such that β is indivisible, (cannot be represented by a multiply covered loop). Then the S^1 equivariant homology $H_*^{S^1}(L_\beta X, \mathbb{Z})$ has finite total rank. Denote by $\chi^{S^1}(L_\beta X)$ the Euler characteristic of this homology. Then*

$$F(g, \beta) = \chi^{S^1}(L_\beta X).$$

Corollary 1.2 above, and its parent Theorem 1.24 lead us to conjecture full topological invariance, see Conjecture 1 in Section 1.1.1. The latter theorem is an important ingredient for some geometric applications concerning closed geodesic counts.

Assuming full topological invariance of our string counting invariant we get the following perhaps mysterious theorem.

Theorem 1.3 (Proof in Section 6)**.** *Assume the Conjecture 1, let $\beta \in \pi_1(T^n)$ be a non-trivial class and let g be a β-regular Finsler metric on T^n having finitely many β class geodesic strings. Suppose that o is a g-geodesic string s.t.:*

(1) *o has class β.*

(2) *$p|\operatorname{mult}(o)$, where p is prime,*

then there is at least one other such o (same p).

Remark 1.4. We can extend the above from metrics to Reeb vector fields on the unit cotangent bundle, representing the standard contact structure. But now counting closed Reeb orbits in classes $\widetilde{\beta}$, lifting a class β as above, (fix a unit speed parametrized representative and lift). However, we need a stronger form of the Conjecture 1, see Remark 1.17.

Without Conjecture 1 we have a local form of the above. Even this local form appears to be inaccessible to standard Morse theory techniques, is a new phenomenon and is likely the most central application of the paper. [1] The main ingredient for this is a product formula for the invariant F in the form of Theorem 8.1. The proof of the latter is rooted in dynamical ideas, but is mostly conceptual, sidestepping intricate analysis sometimes needed for infinite dimensional Morse theory.

Theorem 1.5 (Proof in Section 5)**.** *Let g be the standard flat metric on T^n, and $\beta \in \pi_1(T^n)$ a non-trivial class. For all L sufficiently large, there exists an $\epsilon > 0$ such that whenever g' is a Finsler metric C^2 ϵ-close to g and is β-regular, the following holds. Suppose that o is a g'-geodesic string s.t.:*

(1) *The flat length satisfies $\ell_g(o) < L$.*

(2) *o has class β.*

(3) *$p|\operatorname{mult}(o)$, where p is prime,*

[1] For divisible classes, the Morse-theoretic analogue of the Fuller count would require more than ordinary S^1-equivariant Morse homology. One would want a continuation-invariant Morse theory for the quotient stack $[L_\beta X/S^1]$, together with a compatible rational Euler characteristic which assigns to an isolated nondegenerate critical orbit o the local weight

$$\frac{(-1)^{\operatorname{ind}(o)}}{|\operatorname{Stab}(o)|} = \frac{(-1)^{\operatorname{ind}(o)}}{\operatorname{mult}(o)}.$$

This is the expected orbifold-Euler-characteristic shadow of such a theory, rather than the ordinary Euler characteristic of standard equivariant homology.

then there is at least one other such o (in particular with the same p). Moreover, any Finsler metric g_1 taut deformation equivalent to g (see Definition 1.11) has the same property as g, above. An example for such a metric is in Figure 1.

Recall that a now classical theorem of Preissmann [15] implies that there are no non-trivial compact products with a negative sectional curvature Riemannian metric. The following is one generalization and is one of the main concrete applications of the paper. We denote by $\pi_1^{inc}(X)$ the set of free homotopy classes of loops incompressible to the ends, in the sense of Definition 1.8, (non-constant classes when X is closed).

Theorem 1.6 (Proof in Section 8)**.** *Let Y, Z be smooth manifolds, let $\beta_Z \in \pi_1^{\mathrm{inc}}(Z)$, and assume that $\chi(Y) \neq \pm 1$. Suppose that Y is closed and admits a metric all of whose contractible geodesics are constant, for instance a nonpositive curvature Riemannian metric, and suppose that Z admits a nonpositive curvature metric. Let*

$$X = Y \times Z$$

with projection $X \to Y$, let $i : Z \to X$ be a fiber inclusion, and set

$$\beta = i_*(\beta_Z).$$

Then:

- *X does not admit a complete Riemannian metric of negative sectional curvature.*
- *X does not admit a forward complete Riemann–Finsler metric with a unique and nondegenerate class-β geodesic string.*

This theorem is very close to being sharp, for example the conclusion of the corollary is false if $Y = S^1$ and $Z = S^1 \times \mathbb{R}$. As $X = T^2 \times \mathbb{R}$ admits the warped product metric $g_{T^2} \times_{e^t} g_{\mathbb{R}}$ (with respect to the function $f = e^t$ on $\mathbb{R}$) where $g_{T^2}, g_{\mathbb{R}}$ are the flat metrics. This warped product has constant negative sectional curvature -1. So it is essential that not only $\pi_1(Z)$ be non-trivial but also that there is a class incompressible to the ends. Of course, $\chi(Y) \neq 1$ is also obviously essential, otherwise we may take $Y = pt$. The condition $\chi(Y) \neq -1$ is however not obviously essential.

Remark 1.7. The first part of the theorem above can be proved via the remarkable flat strip theorem [6].

We will also give various generalizations of the above results to fibrations. For fibrations, the most obvious analogue of Preissmann's theorem fails even assuming compactness. In fact, every closed 3-manifold X^3, for which there is no injection $\mathbb{Z}^2 \to \pi_1(X, x_0)$, and which fibers over a circle has a hyperbolic structure g_h, Thurston [17].

1.1. **Setup and more extensive statements.** Proofs of many results here are postponed till subsequent sections.

Terminology 1. From now on, all our metrics are Riemann-Finsler (a.k.a. Finsler) metrics unless specified to be Riemannian, and usually denoted by just g. Completeness, always means forward completeness, and it is an assumption for all our metrics. Curvature always means sectional curvature in the Riemannian case and flag curvature in the Finsler case. Thus we will usually just say complete metric g, for a forward complete Riemann-Finsler metric. A reader may certainly choose to interpret all metrics as Riemannian metrics, completeness as standard completeness, and curvature as sectional curvature.

In what follows, $\pi_k(X)$ denotes the usual based homotopy group, with basepoints suppressed when no confusion can arise. We do not use $\pi_1(X)$ to denote free homotopy classes.

Definition 1.8. *Let X be a smooth manifold. Fix an exhaustion by nested compact sets $\bigcup_{i\in\mathbb{N}} K_i = X$, $K_i \supset K^\circ_{i-1}$ for all $i \geq 1$. A free homotopy class β of loops in X is called* **end compressible** *if, for every i, it has a representative contained in $X - K_i$. We say that β is* **end incompressible** *(or incompressible to the ends) if it is not end compressible.*

Let $\pi_1^{inc}(X)$ denote the set of such end incompressible free homotopy classes. When X is compact, we set $\pi_1^{inc}(X)$ to be the set of nonconstant free homotopy classes of loops in X.

It is easily seen that the above is well defined (independent of the choice of an exhaustion) and moreover any homeomorphism $X_1 \to X_2$ of a pair of manifolds induces a set isomorphism $\pi_1^{inc}(X_1) \to \pi_1^{inc}(X_2)$.

Denote by $L_\beta X$ the class $\beta \in \pi_1^{inc}(X)$ component of the free loop space of X, with its compact open topology. Let g be a complete metric on X, and let $S(g,\beta) \subset L_\beta X$ denote the subspace of all constant speed parametrized, smooth, closed g-geodesics in class β.

Definition 1.9. *We say that a metric g on X is* **β-taut** *if it is complete and $S(g,\beta)$ is compact. We will say that g is* **taut** *if it is β-taut for each $\beta \in \pi_1^{inc}(X)$.*

Lemma 1.10. *A complete metric g with non-positive curvature satisfies:*

- *All of its closed geodesics are minimizing in their free homotopy class.*
- *It is taut.*

Proof. The first part is a standard consequence of the Cartan-Hadamard theorem. The second part follows by the first part and Lemma 4.1. □

It should be emphasized that taut metrics form a much larger class of metrics than just non-positive curvature metrics. For example any sufficiently C^1 small perturbation of a metric with non-positive curvature will be taut. (Indeed, this is crucial for the construction of our invariant.) A simple concrete example very far away from being non-positively curved is in Example (2).

Definition 1.11. *Let $\beta \in \pi_1^{inc}(X)$, and let g_0, g_1 be a pair of β-taut metrics on X. A* **β-taut deformation** *between g_0, g_1, is a continuous (in the topology of C^0 convergence on compact sets) family $\{g_t\}$, $t \in [0,1]$ of complete metrics on X, s.t.*

$$S(\{g_t\},\beta) := \{(o,t) \in L_\beta X \times [0,1] \,|\, o \in S(g_t,\beta)\}$$

is compact. We say that $\{g_t\}$ is a **taut deformation** *if it is β-taut for each $\beta \in \pi_1^{inc}(X)$. The above definitions of tautness are extended naturally to the case of a smooth fibration $X \hookrightarrow P \to [0,1]$, with a smooth fiber-wise family of metrics.*

A useful criterion for β-tautness is the following.

Theorem 1.12 (Proof in Section 2.1)**.** *Let $\beta \in \pi_1^{inc}(X)$. Let $\{g_t\}_{t\in[0,1]}$ be a continuous family of complete metrics on X. Suppose that:*

$$\sup_t |\max_{o\in S(g_t,\beta)} \ell_{g_t}(o) - \min_{o\in S(g_t,\beta)} \ell_{g_t}(o)| < \infty,$$

where ℓ_{g_t} is the length functional with respect to g_t, then $\{g_t\}$ is β-taut.

For example, the hypothesis is trivially satisfied if g_t have the property that all their class β closed geodesics are minimal in their homotopy class.

Corollary 1.13. *If g_t, $t \in [0,1]$ have non-positive curvature then $\{g_t\}$ is taut.*

Proof. This follows by Theorem 1.12 and by Lemma 1.10. □

1.1.1. *The geodesic string counting invariant F.* Let $\mathcal{G}(X)$ be the set of equivalence classes of taut metrics g, where g_0 is equivalent to g_1 whenever there is a taut deformation between them. We may denote an equivalence class by its representative g by a slight abuse of notation. The following is proved in Section 4.

Theorem 1.14. *For each manifold X there is a natural, non-trivial functional:*

$$\mathrm{F} : \mathcal{G}(X) \times \pi_1^{inc}(X) \to \mathbb{Q},$$

extending the definition in (1.1).

For g β-regular and β-taut we have already defined $F(g, \beta)$ at the beginning of the Introduction. For a general β-taut g we define $F(g, \beta)$ in terms of the Fuller index.

Corollary 1.15. *Suppose for a pair g_0, g_1 of β-taut metrics on X:*

$$F(g_0, \beta) \neq F(g_1, \beta),$$

then any path $\{g_t\}$, connecting g_0, g_1, is not β-taut.

Proof. The fact that any connecting $\{g_t\}$ is not β-taut is just a direct corollary of the theorem above. □

I was unable to find such a pair g_0, g_1, in fact there is strong evidence to believe the following:

Conjecture 1. *F does not depend on the choice of a smooth structure and taut metric. In other words we have the following. Let X be a topological manifold, define:*

$$\mathcal{S}(X) : \pi_1^{inc}(X) \to \mathbb{Q} \sqcup \{\infty\}$$

by:

$$\mathcal{S}(X)(\beta) = \begin{cases} \infty, & \text{if } X \text{ does not admit a smooth structure and a } \beta\text{-taut Finsler metric.} \\ F(g, \beta), & \text{if } g \text{ is a } \beta\text{-taut Finsler metric on } X. \end{cases},$$

then $\mathcal{S}$ is well defined and hence determines a topological invariant of topological manifolds. So if $f : X_1 \to X_2$ is a homeomorphism then $\mathcal{S}(X_1)(\beta) = \mathcal{S}(X_2)(f_*(\beta))$.

Theorem 1.24 in the following section partially proves this conjecture.

Remark 1.16. The smooth manifold version of this conjecture, i.e. that $\mathcal{S}(X)$ is a smooth manifold invariant, is implied by the Reeb sky catastrophe conjecture described in the next section. However, the above seems to be much more basic as will be apparent from the proof of Theorem 1.24.

Remark 1.17. The formulation of the conjecture naturally extends to Reeb vector fields. That is let λ be a contact form on the unit cotangent bundle C of a smooth manifold X (for simplicity closed), with λ representing the Liouville contact structure (i.e. the standard contact structure). Assume that the space of closed, class $\widetilde{\beta}$ λ-Reeb orbits on C is compact. Then the Fuller index (See Section 4) of this compact set is a topological invariant of X.

1.1.2. *A short digression on sky catastrophes.*

Definition 1.18 (Preliminary). *A* **sky catastrophe** *for a smooth family $\{X_t\}$, $t \in [0,1]$, of non-vanishing vector fields on a closed manifold M is a continuous family of closed orbit strings $\tau \mapsto o_{t_\tau}$, o_{t_τ} is an orbit string of X_{t_τ}, $\tau \in [0,\infty)$, such that the period of o_{t_τ} is unbounded from above.*

Sky catastrophes first appeared in the work of Fuller [7]. One general definition appears in [16], which we review in the Appendix C, the main point of this definition is that there are no regularity assumptions on $\{X_t\}$. The preliminary definition becomes equivalent to the more general definition given certain regularity conditions on the family $\{X_t\}$.

The following corollary of Theorem 1.12 may be of independent interest.

Corollary 1.19. *Sky catastrophes of vector fields on closed manifolds are not geodesible by metrics all of whose geodesics are minimal. That is if a family $\{X_t\}$ has a sky catastrophe, there is no family $\{g_t\}$ of metrics such that the orbits of X_t are unit speed parametrized g_t-geodesics.*

Fuller at the end of [8] has asked for any metric conditions on vector fields to rule out sky catastrophes. By the above, non-positivity of curvature is one such condition. So this is a partial answer to his question.

The main conjecture in [16] is that geodesible, and more generally Reeb sky catastrophes ($\{X_t\}$ are Reeb) are unstable, in the sense that there is a C^0 nearby family $\{X'_t\}$ which does not have a sky catastrophe (at least in the fixed class β). Let's call this ***Reeb sky catastrophe conjecture***. The reason to conjecture such a thing, is that a theorem in [16] says that if $\{o_{t_\tau}\}_\tau$ is a sky catastrophe for a family of Reeb vector fields $\{X_t\}$ then the length of the corresponding projection $\tau \mapsto t_\tau$ is infinite. (Assuming some minor regularity on $\{o_{t_\tau}\}$.) The latter suggests that the structure of such a sky catastrophe is very fragile.

Remark 1.20. The Reeb sky catastrophe conjecture implies the geodesible Seifert conjecture (apparently still an open problem), by the main result of [16]. Hence, this is a subtle question.

The following is a variant of Corollary 1.15.

Theorem 1.21. *Suppose for a pair g_0, g_1 of β-taut metrics on a closed manifold M:*

$$F(g_0,\beta) \neq F(g_1,\beta),$$

then any path $\{g_t\}$, connecting g_0, g_1, has a sky catastrophe, meaning that the corresponding family $\{X_t\}$ of Reeb vector fields on the unit cotangent bundle (cf. Section 4) has a sky catastrophe as in Definition C.1.

Proof. This directly follows by [16, Theorem 3.2]. □

So that if such a pair g_0, g_1 exists the Reeb sky catastrophe conjecture is disproved, and in a rather extreme fashion.

1.1.3. *Basic results on the invariant F.*

Definition 1.22. *Let β be a free homotopy class of loops in X. Choose a representative of β and a base point x_0 on its image. This choice determines an element*

$$\beta_{x_0} \in \pi_1(X, x_0)$$

well-defined up to inner automorphism.

- *An element $\gamma \in \pi_1(X, x_0)$ is* **indivisible** *if whenever $\gamma = \alpha^k$ for some $\alpha \in \pi_1(X, x_0)$ and $k > 0$, then $k = 1$.*
- *An element $\gamma \in \pi_1(X, x_0)$ is a k-***power** *if $\gamma = \alpha^k$, $k > 1$, for some α which is not an n-power for any $n > 1$.*
- *An element $\gamma \in \pi_1(X, x_0)$ is* **atomic** *if it is a k-power for some $k > 1$.*

We say that the free homotopy class β is indivisible, respectively a k-power, respectively atomic, if β_{x_0} has the corresponding property for one, equivalently any, such choice of x_0 and representative. This is well-defined because the corresponding notions are invariant under inner automorphism. Equivalently, β is a k-power precisely when it admits a smooth representative with multiplicity k whose underlying primitive class is not itself a proper power. [2]

Note that a class $\beta \in \pi_1^{inc}(X)$ is indivisible iff any representative of this class is not multiply covered.

Example 1. Let g be a Riemannian metric with negative sectional curvature on a closed manifold X and $\beta \in \pi_1^{inc}(X)$ a class represented by a multiplicity n closed geodesic, then

$$F(g, \beta) = \frac{1}{n}. \tag{1.23}$$

In particular, if β is indivisible then $F(g, \beta) = 1$. More generally, (1.23) holds whenever g has a unique and non-degenerate geodesic string in class β, where non-degenerate is as in the Introduction.

If $\beta \in \pi_1^{inc}(X)$ is indivisible, then it is easy to see that the reparametrization S^1 action on $L_\beta X$ is free (Section 7), so that $H_*^{S^1}(L_\beta X, \mathbb{Z}) \simeq H_*(L_\beta X/S^1, \mathbb{Z})$, where $H_*^{S^1}(L_\beta X, \mathbb{Z})$ denotes the S^1-equivariant homology. Moreover, we have the following result proved in Section 7.

Theorem 1.24. *Suppose that $\beta \in \pi_1^{inc}(X)$ is indivisible, and X admits a β-taut metric. Then $H_*^{S^1}(L_\beta X, \mathbb{Z})$ has finite total rank. Denote by $\chi^{S^1}(L_\beta X)$ the Euler characteristic of this homology. Then for any β-taut metric g on X:*

$$F(g, \beta) = \chi^{S^1}(L_\beta X).$$

Consequently, $F(g, \beta)$ is independent of the choice of β-taut metric g, for indivisible β. And Furthermore, Conjecture 1 holds on the subset of classes β which are indivisible.

Example 2. A basic example of a β-taut metric that is not nearby to a non-positive curvature metric is the following. Let g be a metric on the 2-torus with exactly two class β g-geodesic strings, for β one of the π_1 generators. We can take such a g_0 with arbitrarily large curvature, see Figure 1. The metric g_0 in this figure is moreover clearly taut deformation equivalent to the standard flat metric. Taking products of metrics of this sort gives higher dimensional examples.

More examples for the theorem above can be found by the proof of Theorem 1.26.

Remark 1.25. If β is a not indivisible, the idea behind Theorem 1.24 breaks down, as the S^1-equivariant homology of $L_\beta X$ may then be infinite dimensional even if X admits a β-taut g. As a trivial example, this homology is already infinite dimensional when g is negatively curved, and the class β geodesic is k-covered, as then this homology is the group homology of $\mathbb{Z}_k$. For general β, one expects instead a Morse theory for the quotient stack $[L_\beta X/S^1]$, together with a compatible rational orbifold Euler characteristic. A nondegenerate string o should contribute

$$\frac{(-1)^{\text{morse}(o)}}{\text{mult}(o)}.$$

[2]As we are working with non-compact manifolds, infinitely divisible free homotopy classes may occur in principle.

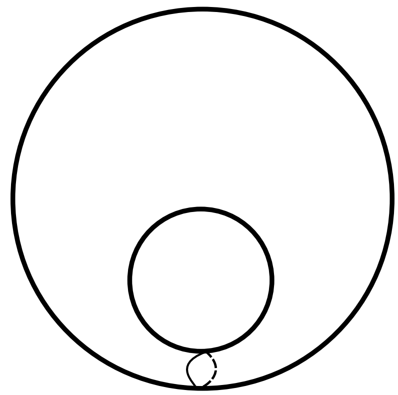

FIGURE 1. The class β is represented by the drawn curve.

This weighted Euler characteristic would recover the Fuller count. We do not construct such a theory here.

We can use the above, and the product formula of Theorem 8.1 to get:

Theorem 1.26 (Proof in Section 8)**.** *Every rational number has the form $F(g, \beta)$ for some β-taut Riemannian g on some compact manifold X and for some β.*

1.1.4. *Applications to existence of negative curvature metrics.* A celebrated theorem of Preissmann [15] says that there are no negative sectional curvature metrics on compact products. Fibration counterexamples to Preissmann's product theorem certainly exist as mentioned in the Introduction. We are going to give a certain generalization of Preissmann's theorem to fibrations, with possibly non-compact fibers, also replacing the negative sectional curvature condition by a significantly weaker condition. Though the definitions may at first seem hard to understand, we just want enough generality to incorporate interesting "flat bundle" examples (with anti-Anosov or periodic holonomy behavior). The latter are of interest in 3-manifold theory, for instance vis-à-vis Thurston's classification [17].

Definition 1.27. *For a fiber bundle $Z \hookrightarrow X \to Y$, we say that $\beta \in \pi_1^{\mathrm{inc}}(X)$ is a* **fiber class** *if it lies in the image of the set map*
$$i_Z : \pi_1^{\mathrm{inc}}(Z) \to \pi_1^{\mathrm{inc}}(X)$$
induced by inclusion of a fiber.

Lemma 1.28. *Let*
$$Z \hookrightarrow X \xrightarrow{p} Y$$
be a smooth fiber bundle with connected fiber and connected base. Assume
$$\pi_2(Y) = 0.$$
Fix $y_0 \in Y$, and let
$$H = \pi_1(Y, y_0)$$
act on $\pi_1^{\mathrm{inc}}(Z_{y_0})$ by holonomy. Let $\beta \in \pi_1^{\mathrm{inc}}(X)$ be a fiber class, and define
$$\mathcal{R}_{y_0}(\beta) = \left\{ \alpha \in \pi_1^{\mathrm{inc}}(Z_{y_0}) \;\middle|\; (i_{y_0})_*(\alpha) = \beta \right\}.$$
Then $\mathcal{R}_{y_0}(\beta)$ is a single H-orbit. In particular, if
$$\beta_Z, \beta'_Z \in \mathcal{R}_{y_0}(\beta),$$
then
$$H \cdot \beta_Z = H \cdot \beta'_Z.$$

Thus the holonomy orbit associated to the fiber class β is independent of the choice of fiber representative β_Z.

Proof. Choose basepoints $z_0 \in Z_{y_0}$ and $x_0 = i_{y_0}(z_0)$. Since $\pi_2(Y) = 0$, the homotopy long exact sequence gives an exact sequence of based groups

$$1 \to \pi_1(Z_{y_0}, z_0) \xrightarrow{(i_{y_0})_*} \pi_1(X, x_0) \xrightarrow{p_*} \pi_1(Y, y_0) \to 1.$$

Thus we identify $\pi_1(Z_{y_0}, z_0)$ with a normal subgroup of $\pi_1(X, x_0)$, and the quotient is $\pi_1(Y, y_0)$.

Let $\alpha, \alpha' \in \mathcal{R}_{y_0}(\beta)$. Choose based representatives of these fiber free homotopy classes. Since they determine the same total free homotopy class β in X, the corresponding elements of $\pi_1(X, x_0)$ are conjugate. Thus, after conjugating inside $\pi_1(Z_{y_0}, z_0)$ if necessary, there exists $x \in \pi_1(X, x_0)$ such that

$$\alpha' = x\alpha x^{-1}.$$

Let

$$h = p_*(x) \in H.$$

Conjugation by x on the normal subgroup $\pi_1(Z_{y_0}, z_0)$ induces the holonomy action of h, up to an inner automorphism of $\pi_1(Z_{y_0}, z_0)$. Passing to free homotopy classes in the fiber, we get

$$\alpha' = h \cdot \alpha.$$

Thus all fiber representatives of the total free homotopy class β belong to a single holonomy orbit.

Conversely, if $\alpha' = h \cdot \alpha$, choose $x \in \pi_1(X, x_0)$ with $p_*(x) = h$. Then $x\alpha x^{-1}$ represents the same total free homotopy class in X as α, and its fiber free homotopy class is $h \cdot \alpha$. Hence the holonomy orbit is exactly the set of fiber representatives of β. ☐

For a fiber class $\beta \in \pi_1^{\mathrm{inc}}(X)$, set

$$\mathcal{R}(\beta) = \left\{\alpha \in \pi_1^{\mathrm{inc}}(Z) \,\middle|\, (i_Z)_*(\alpha) = \beta\right\}.$$

Assume Y is connected and $\pi_2(Y) = 0$, and put

$$H = \pi_1(Y, y_0).$$

By Lemma 1.28, the holonomy orbit

$$S = H \cdot \beta_Z$$

is independent of the choice of the fiber representative $\beta_Z \in \mathcal{R}(\beta)$. If another fiber inclusion is chosen, the corresponding holonomy actions are abstractly conjugate.

Definition 1.29. *Let*

$$Z \hookrightarrow X \xrightarrow{p} Y$$

*be a smooth fiber bundle, with Y connected and $\pi_2(Y) = 0$, and let $\beta \in \pi_1^{\mathrm{inc}}(X)$ be a fiber class. We say that p is β-**holonomy-finite** if, for one, equivalently any, $\beta_Z \in \mathcal{R}(\beta)$, the orbit*

$$S = \pi_1(Y, y_0) \cdot \beta_Z$$

is finite.

Definition 1.30. *Let*

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a smooth fiber bundle, and let g be a Riemannian metric on X. Let

$$T^{\mathrm{vert}}X = \ker(dp)$$

and let

$$\pi^{\mathrm{vert}} : TX \to T^{\mathrm{vert}}X$$

*be the g-orthogonal projection. We say that p is g-***parallel** *if*

$$\nabla \pi^{\mathrm{vert}} = 0.$$

Equivalently,

$$\nabla_A(\pi^{\mathrm{vert}} B) = \pi^{\mathrm{vert}}(\nabla_A B)$$

for all vector fields A, B on X.

Definition 1.31. *Let $\beta \in \pi_1^{\mathrm{inc}}(X)$ be a fiber class, for a smooth fibration*

$$p : Z \hookrightarrow (X, g) \to (Y, g_Y).$$

*We say that this is a β-***periodic fibration** *if:*

1. *g is a complete Riemannian metric.*
2. *Y is connected and $\pi_2(Y) = 0$.*
3. *p is g-parallel and is a Riemannian submersion.*
4. *p is β-holonomy-finite.*
5. *the fiber metrics g_y on Z_y are taut.*

The metrics g, g_Y may be kept implicit. In the above definition of a β-periodic fibration and the hypothesis of the following theorem we need the auxiliary metric g on X to be Riemannian, and there is no obvious extension of the theorem to the Riemann-Finsler case. However, the conclusions of the theorem are for Riemann-Finsler metrics. The following is perhaps the main technical result of the paper, proved in Section 8.

Theorem 1.32. *Let $p : Z \hookrightarrow (X, g) \to (Y, g_Y)$ be a β-periodic fibration, where $\beta \in \pi_1^{inc}(X)$ is a fiber class. Suppose further that Y is connected, closed, $\chi(Y) \neq \pm 1$ and is such that all smooth closed contractible g_Y-geodesics in Y are constant. Then the following holds:*

- *X does not admit a complete Riemannian metric with negative curvature.*
- *X does not admit a forward complete Riemann-Finsler metric with a unique nondegenerate class-β geodesic string.*

Note that $\chi(Y) \neq 1$ is of course essential, as the trivial fibration $X \to \{pt\}$, with X admitting a complete negatively curved metric, will satisfy the hypothesis. The condition that there is a fiber class $\beta \in \pi_1^{inc}(X)$ is also essential, for any vector bundle over a manifold admitting a Riemannian metric of negative curvature admits a metric of negative curvature, Anderson [1].

Theorem 1.6 gives one set of examples. A further basic set of examples for the theorem is obtained by starting with a Riemannian manifold (Y, g_Y), and any homomorphism

$$\phi : \pi_1(Y, y_0) \to \mathrm{Isom}(Z, g_Z), \quad \text{(the group of all isometries).} \tag{1.33}$$

where g_Z is a taut Riemannian metric and $\beta_Z \in \pi_1^{inc}(Z)$. (For example (Z, g_Z) is a non-simply connected complete hyperbolic surface of genus at least one). Suppose further:

1. The orbit
$$Orb := \bigcup_{\gamma \in \pi_1(Y, y_0)} \phi_*(\gamma)(\beta_Z)$$
is finite.
2. Y is closed and connected.
3. All contractible smooth closed g_Y geodesics in Y are constant.

We have the obvious induced diagonal action

$$\pi_1(Y, y_0) \to \mathrm{Diff}(Z \times \widetilde{Y}), \text{ (the group of all diffeomorphisms)},$$

$$\gamma \mapsto ((z, y) \mapsto (\phi(\gamma)(z), \gamma \cdot y)),$$

for $\widetilde{Y}$ the universal cover of Y. Taking the quotient of $Z \times \widetilde{Y}$ by this action, we get an associated "flat" bundle $Z \hookrightarrow X_\phi \xrightarrow{p} Y$, with a metric g_ϕ induced from the product metric $\widetilde{g} = g_Z \oplus g_Y$, on the covering space $q : Z \times \widetilde{Y} \to Z \times Y$. The natural projection $p : X_\phi \to Y$ is then a Riemannian submersion. In particular p is a β-periodic submersion. Hence $p : (X_\phi, g_\phi) \to (Y, g_Y)$ satisfies the hypothesis of Theorem 1.32. Yet more concretely:

Example 3. Suppose we have $\beta_Z \in \pi_1^{inc}(Z)$, and let $\phi : Z \to Z$ be an isometry of a taut metric g_Z, such that *Orb* is finite. Then for $\beta = (i_Z)_*(\beta_Z)$, by the construction above, the mapping torus

$$(Z, g_Z) \hookrightarrow (X_\phi, g_\phi) \xrightarrow{\pi} S^1$$

has the structure of a β-periodic fibration, satisfying the hypothesis of Theorem 1.32.

Corollary 1.34. *Let*

$$(Z, g_Z) \hookrightarrow (X_\phi, g_\phi) \to (Y, g_Y)$$

be as in the construction above for Z, g_Z having non-positive curvature, and such that Orb is finite. (If Z is closed then the finiteness is automatic, since $\mathrm{Isom}(Z, g_Z)$ *is a compact Lie group and its identity component acts trivially on free homotopy classes.) Let $\beta_Z \in \pi_1^{inc}(Z)$. Then if $\chi(Y) \neq \pm 1$:*

1. *X_ϕ does not admit a complete Riemannian metric with negative curvature.*
2. *Moreover, X_ϕ does not admit a Riemann-Finsler metric with a unique nondegenerate class-β geodesic string, for $\beta = i_*(\beta_Z)$ as above.*

As a special case, this applies to the mapping tori X_ϕ, for $\phi : Z \to Z$ an isometry of a complete Riemannian non-positively curved metric on Z, satisfying the finiteness condition 1. (The non-positive curvature hypothesis is for concreteness we may of course replace this condition by tautness.)

In the special case when Z is compact, the first part of the above corollary can likely be deduced, from Preissmann's theorem, see also [3, Theorem 9.3.4] for a generalization that fits our Finsler setting. The second part is new even when Z is compact.

The following questions seem to be open.

Question 1. Is the β-holonomy finite assumption necessary for Corollary 1.34?

Question 2. Let Z be an infinite type surface, can (X_ϕ, g_ϕ) have a complete negative curvature metric? The corollary above cannot give a negative answer since in this case $X_\phi \to S^1$ may not be β-holonomy finite. But the answer is very likely no, as the intuition of Thurston-Nielsen classification (it cannot be directly applied in the infinite type setting) suggests that ϕ should have pseudo-Anosov type behavior which an isometry even on an infinite type surface cannot have.

2. Compactness preliminaries

Lemma 2.1. *Suppose that g is a complete metric on X, $\beta \in \pi_1^{inc}(X)$ and let $S \subset L_\beta X$ be a subset on which the g-length functional is bounded from above. Then the images in X of elements of S are contained in a fixed compact subset of X.*

Proof. Suppose otherwise. Fix an exhaustion by nested compact sets

$$\bigcup_{i\in\mathbb{N}} K_i = X, \quad K_i^\circ \supset K_{i-1}.$$

Then either there is sequence $\{o_i\}_{i\in\mathbb{N}}$, $o_i \in S$ s.t. $o_i \in K_i^c$, for K_i^c the complement of K_i, which contradicts the fact that β is end incompressible. Or there is a sequence $\{o_k\}_{k\in\mathbb{N}}$, $o_k \in S$ s.t.:

(1) Each o_k intersects K_{i_0} for some i_0 fixed.

(2) For each $i \in \mathbb{N}$ there is a $k_i > i$ s.t. o_{k_i} is not contained in K_i.

Now if $\operatorname{diam}(o_k)$ is bounded in k, then condition 1 implies that o_k are contained in a set of bounded diameter. (Here $\operatorname{diam}(o_k)$ denotes the diameter of image o_k.) Consequently, by Hopf-Rinow theorem [2], o_k are contained in a compact set. But this contradicts condition 2, and the fact that K_i form an exhaustion of X.

Thus, we conclude that $\operatorname{diam}(o_k)$ is unbounded, but this contradicts the hypothesis. □

Lemma 2.2. *Let g be a complete metric on X, and let $\beta \in \pi_1^{\mathrm{inc}}(X)$. Denote by $S_A(g,\beta) \subset S(g,\beta)$ the subset of elements with g-length at most A. Then $S_A(g,\beta) \subset S(g,\beta)$ is compact.*

Proof. By Lemma 2.1, the images of elements of $S_A(g,\beta) \subset S(g,\beta)$ are contained in compact K. Since elements of $S_A(g,\beta) \subset S(g,\beta)$ are parametrized by constant speed, and since we have a length bound $S_A(g,\beta)$ is an equicontinuous collection. The result then follows by the Arzelà-Ascoli theorem. (This uses the standard fact that C^0 convergence of closed geodesics upgrades to C^∞ convergence provided g is smooth.) □

Lemma 2.3. *Let g be a complete metric on X, and let $\beta \in \pi_1^{\mathrm{inc}}(X)$. Then there exists a closed g-geodesic minimizing length among all loops in the free homotopy class β.*

Proof. Suppose otherwise and let

$$A = \inf_{\gamma\in L_\beta X} \ell_g(\gamma).$$

By the hypothesis that there is no geodesic with length A, by Lemma 2.2, there is an $\epsilon > 0$ such that there is no closed geodesic o with $A < \ell_g(o) < A + \epsilon$.

By Lemma 2.1 the subset $M_{\beta,A} \subset L_\beta X$ consisting of elements with length less than $A+\epsilon$ is contained in a compact K. Pick $o \in M_{\beta,A}$ and apply the curve shortening process to it. Let o_k, $k \in \mathbb{N}$ be the stages of this process. Then image $o_k \subset K$. In particular $\{o_k\}$ must converge to a geodesic [13], but this contradicts the condition on ϵ. □

2.1. **Proof of Theorem 1.12.** Let $\{g_t\}$, $t \in [0,1]$ be as in the hypothesis, with

$$\sup_t \left| \max_{o\in S(g_t,\beta)} \ell_{g_t}(o) - \min_{o\in S(g_t,\beta)} \ell_{g_t}(o) \right| < C, \tag{2.4}$$

and suppose that

$$\sup_{(o,t)\in\mathcal{O}(\{g_t\},\beta)} \ell_{g_t}(o) = \infty.$$

Then we have a sequence $\{o_k\}$, $k \in \mathbb{N}$, of closed class β g_{t_k}-geodesics in X, satisfying:

(1) $\lim_{k\to\infty} t_k = t_\infty \in [0,1]$.

(2) $\lim_{k\to\infty} \ell_{g_{t_k}}(o_k) = \infty$, where $\ell_{g_{t_k}}(o_{t_k})$ is the length with respect to g_{t_k}.

Applying Lemma 2.3, we may choose a length-minimizing closed g_{t_∞}-geodesic o_∞ in the class β. And let L denote its length g_∞ length. Let g_{aux} be a fixed auxiliary metric on X, and let L_{aux} be the g_{aux} length of o_∞.

Define a pseudo-metric d_{C_0} on the space of metrics on X as follows. Set $K = \operatorname{image} o_\infty$. And set

$$V \subset TX = \{v \in TX \mid \pi(v) \in K \text{ for } \pi : TX \to X \text{ the canonical projection, and } |v|_{aux} = 1\},$$

where $|v|_{aux}$ is the norm taken with respect to g_{aux}.

Then define:

$$d_{C^0}(g_1, g_2) = \sup_{v \in V} ||v|_{g_1} - |v|_{g_2}|.$$

By Properties 1 and 2 we may find a $k > 0$ such that:

$$d_{C^0}(g_{t_k}, g_{t_\infty}) < \epsilon \tag{2.5}$$

and

$$\ell_{g_{t_k}}(o_k) > C + L + L_{aux} \cdot \epsilon. \tag{2.6}$$

By (2.5), we have:

$$\ell_{g_{t_k}}(o_\infty) < \ell_{g_{t_\infty}}(o_\infty) + L_{aux} \cdot \epsilon = L + L_{aux} \cdot \epsilon.$$

Combining with (2.6) we get:

$$\ell_{g_{t_k}}(o_k) > \ell_{g_{t_k}}(o_\infty) + C.$$

Since we may find a closed g_{t_k}-geodesic o' satisfying $\ell_{g_{t_k}}(o') \leq \ell_{g_{t_k}}(o_\infty)$, we get that

$$|\max_{o \in S(g_{t_k}, \beta)} \ell_{g_{t_k}}(o) - \min_{o \in S(g_{t_k}, \beta)} \ell_{g_{t_k}}(o)| > C,$$

and so we are in contradiction.

Thus,

$$\sup_{(o,t) \in \mathcal{O}(\{g_t\}, \beta)} \ell_{g_t}(o) < \infty.$$

Then compactness of $\mathcal{O}(\{g_t\}, \beta)$ follows by a direct analogue of Lemma 2.2. □

3. Preliminaries on Reeb flow

Let (C^{2n+1}, λ) be a contact manifold with λ a contact form, that is a one form s.t. $\lambda \wedge (d\lambda)^n \neq 0$. Denote by R^λ the Reeb vector field satisfying:

$$d\lambda(R^\lambda, \cdot) = 0, \quad \lambda(R^\lambda) = 1.$$

Recall that a ***closed λ-Reeb orbit*** (or just Reeb orbit when λ is implicit) is a smooth map

$$o : (S^1 = \mathbb{R}/\mathbb{Z}) \to C$$

such that

$$\dot{o}(t) = cR^\lambda(o(t)),$$

with $\dot{o}(t)$ denoting the time derivative, for some $c > 0$ called period. Let $S(R^\lambda, \beta)$ denote the space of all closed λ-Reeb orbits in free homotopy class β, with its compact open topology. And set

$$\mathcal{O}(R^\lambda, \beta) = S(R^\lambda, \beta)/S^1,$$

where $S^1 = \mathbb{R}/\mathbb{Z}$ acts by reparametrization $t \cdot o(\tau) = o(t + \tau)$.

4. Definition of the functional F and proofs of auxiliary results

Let X be a manifold. Fix once and for all an auxiliary Riemannian metric g_0, and set

$$C = S^*_{g_0} X.$$

For any Riemann–Finsler metric g, let F^*_g denote the dual norm, and let

$$\rho_g : C \to S^*_g X, \qquad \rho_g(\xi) = \frac{\xi}{F^*_g(\xi)}$$

be the fiberwise radial projection. We define a contact form on the fixed manifold C by

$$\lambda_g = \rho_g^* \lambda_{\mathrm{can}},$$

where λ_{can} is the Liouville one-form on T^*X. We write

$$R^{\lambda_g}$$

for the Reeb vector field of λ_g on this fixed manifold C. Under ρ_g, the closed orbits of R^{λ_g} correspond to the usual closed g-geodesics on $S^*_g X$.

The same convention applies to the Sasaki metric. If g is Riemannian, let g_S^{nat} denote the usual Sasaki metric on $S^*_g X$, and set

$$g_S^g = \rho_g^* g_S^{\mathrm{nat}}$$

on the fixed manifold C. When only one metric is under discussion, we write g_S for g_S^g. Similarly, all metric-dependent functions on $S^*_g X$, such as verticality functions, are regarded as functions on C by pullback through ρ_g. This fixed-C convention will be used throughout.

Suppose now that g is β-taut. If

$$o : S^1 = \mathbb{R}/\mathbb{Z} \to X$$

is a nonconstant constant-speed closed g-geodesic, it has a canonical unit cotangent lift to $S^*_g X$, and hence, using ρ_g^{-1}, to the fixed manifold C. Let

$$\widetilde{\beta}$$

be the free homotopy class of such a lift. This class is independent of the chosen class-β closed geodesic in the compact block under consideration.

Let

$$\mathcal{O}_{g,\beta} = \mathcal{O}(R^{\lambda_g}, \widetilde{\beta})$$

be the corresponding orbit-string space. By construction, $\mathcal{O}_{g,\beta}$ is identified with the space of class-β g-geodesic strings, and by β-tautness it is compact. We define

$$F(g, \beta) = i(\mathcal{O}_{g,\beta}, R^{\lambda_g}, \widetilde{\beta}) \in \mathbb{Q},$$

where the right hand side is the Fuller index as outlined in Appendix A. As a basic example we have:

Lemma 4.1. *Suppose that g is a complete metric on X, all of whose class $\beta \in \pi_1^{inc}(X)$ geodesics are minimal, then g is β-taut.*

Proof. This is a corollary of Lemma 2.2. □

Proof Theorem 1.14. Let $\beta \in \pi_1^{inc}(X)$ be given, and let g be β-taut. We just need to prove that $F(g, \beta)$ is invariant under a β-taut deformation of g. Let $\{g_t\}_{t \in [0,1]}$ be a β-taut deformation of metrics. Using the fixed-C convention above, each metric g_t determines a contact form λ_{g_t} and a Reeb vector field $R^{\lambda_{g_t}}$ on the same manifold C. Let

$$\mathcal{O}(\{R^{\lambda_{g_t}}\}, \widetilde{\beta})$$

be the orbit cobordism as in (A.1). This cobordism is compact because $S(\{g_t\}, \beta)$ is compact by the definition of a β-taut deformation. Basic invariance of the Fuller index, that is (A.2), immediately yields: $F(g_0, \beta) = F(g_1, \beta)$. □

5. Proof of Theorem 1.5

Let β be a non-zero class, as in the statement. Let $Y \subset T^n$ be a totally g-geodesic submanifold diffeomorphic to T^{n-1}, containing image β. Let $\alpha \in H^1(T^n, \mathbb{Z})$ be the Poincare dual of Y and let $p : T^n \to S^1$ be the classifying map of α. Then clearly p is a β-periodic fibration with respect to g.

We work on the fixed contact manifold

$$C = S^*_g T^n,$$

and let h be a fixed auxiliary metric on C, for instance the Sasaki metric associated to the flat metric g. The nearby unit cotangent bundles are identified with C by the fixed radial convention of Section 4; hence $R^{\lambda_{g'}}$ denotes the corresponding pushed-forward Reeb vector field on C.

Let L_β denote the flat length of a class-β geodesic, and let $L > L_\beta$ be the length cutoff appearing in the statement. Since C is compact and $R^{\lambda_{g'}} \to R^{\lambda_g}$ in C^0 for g' sufficiently C^2-close to g, there is a preliminary C^2-neighborhood of g and constants

$$0 < a < A < \infty$$

such that, for every metric g'' in this neighborhood,

$$|R^{\lambda_{g''}}|_h \leq A, \qquad |\operatorname{pr}_* R^{\lambda_{g''}}|_g \geq a$$

on C. Set

$$A_L = \left(\frac{A}{a} + 1\right) L.$$

Define

$$\widehat{U}_{L,A_L} = \left\{ \gamma \in \mathcal{L}_{\widetilde{\beta}} C \,\middle|\, \ell_g(\operatorname{pr} \circ \gamma) < L, \quad \ell_h(\gamma) < A_L \right\},$$

and let

$$U_{L,A_L} = \widehat{U}_{L,A_L} / S^1$$

be its image in the orbit-string quotient. This is a controlled neighborhood of

$$N_0 = \mathcal{O}(R^{\lambda_g}, \widetilde{\beta}).$$

Indeed, the h-length bound gives the control required in Lemma A.3, and $L > L_\beta$ ensures that $N_0 \subset U_{L,A_L}$. Since the flat metric has no other class-β geodesic strings, U_{L,A_L} is isolating for N_0.

Let W be as in Lemma A.3, and let ϵ be smaller than both the corresponding Fuller constant for

$$(N_0, U_{L,A_L}, W)$$

and the size of the preliminary C^2-neighborhood above. If g' is $C^2 \epsilon$-close to g, Lemma A.4 gives

$$i(N_0, R^{\lambda_g}, \widetilde{\beta}) = i(N_1, R^{\lambda_{g'}}, \widetilde{\beta}), \tag{5.1}$$

where

$$N_1 = U_{L,A_L} \cap \mathcal{O}(R^{\lambda_{g'}}, \widetilde{\beta}).$$

We now identify N_1 with the length-filtered class-β geodesic strings of g'. Let γ be a closed $R^{\lambda_{g'}}$-orbit, parametrized by

$$\dot{\gamma} = T R^{\lambda_{g'}}(\gamma).$$

Then

$$\ell_h(\gamma) \leq AT,$$

while

$$\ell_g(\mathrm{pr}\circ\gamma) = T\int_0^1 |\,\mathrm{pr}_* R^{\lambda_{g'}}(\gamma(t))|_g\, dt \geq aT.$$

Thus

$$\ell_h(\gamma) \leq \frac{A}{a}\,\ell_g(\mathrm{pr}\circ\gamma).$$

Consequently, for closed $R^{\lambda_{g'}}$-orbits,

$$\ell_g(\mathrm{pr}\circ\gamma) < L \implies \ell_h(\gamma) < A_L.$$

Since the definition of U_{L,A_L} already includes the condition $\ell_g(\mathrm{pr}\circ\gamma) < L$, the set N_1 is exactly the set of class-β g'-geodesic strings satisfying $\ell_g(o) < L$.

By Theorem 8.1, $F(g,\beta) = 0$, since $\chi(S^1) = 0$. Also the elements of N_1 are in correspondence with elements of

$$\mathcal{O}_L(g',\beta) = \{o \in \mathcal{O}(g',\beta) \,|\, \ell_g(o) < L\}.$$

We may thus rewrite (5.1) as:

$$\sum_{o\in\mathcal{O}_L(g',\beta)} \frac{(-1)^{\mathrm{morse}(o)}}{\mathrm{mult}(o)} = 0, \tag{5.2}$$

Since g' is β-regular and N_1 is compact, $\mathcal{O}_L(g',\beta)$ is finite. Enumerate its elements as $o_1,\ldots,o_n$. We then get:

$$\sum_i (-1)^{\mathrm{morse}(o_i)} \prod_{j\neq i} \mathrm{mult}(o_j) = 0.$$

We show that if a prime p divides the multiplicity $\mathrm{mult}(o_k)$ for some $k \in \{1,\ldots,n\}$, then there exists at least one other index $m \neq k$ such that p divides $\mathrm{mult}(o_m)$. This will complete the proof.

Let $M_i = \prod_{j\neq i}\mathrm{mult}(o_j)$ and let $\sigma_i = (-1)^{\mathrm{morse}(o_i)}$. The original equation is:

$$\sum_{i=1}^n \sigma_i M_i = 0.$$

Suppose $p \mid \mathrm{mult}(o_k)$. Then

$$\sum_{i=1}^n \sigma_i M_i \equiv 0 \pmod p. \tag{5.3}$$

For any index $i \neq k$, the product M_i contains the factor $\mathrm{mult}(o_k)$:

$$M_i = \mathrm{mult}(o_k)\cdot \prod_{j\neq i,k}\mathrm{mult}(o_j).$$

Since $p \mid \mathrm{mult}(o_k)$, it follows from (5.3) that:

$$\sigma_k M_k + 0 \equiv 0 \pmod p.$$

And so

$$M_k \equiv 0 \pmod p.$$

By definition, $M_k = \prod_{j\neq k}\mathrm{mult}(o_j)$. Since p is prime and $p \mid \prod_{j\neq k}\mathrm{mult}(o_j)$, by Euclid's Lemma,

$$p \mid \mathrm{mult}(o_j) \text{ for some } j \neq k.$$

And this completes the proof of the claim and the main theorem.

$\square$

6. Proof of Theorem 1.3

Let $p : T^n \to S^1$ be the β-periodic fibration with respect to g_0, as in the proof of Theorem 1.5 above. By Theorem 8.1, $F(g_0, \beta) = 0$. Hence, as we assumed Conjecture 1, for any other β-regular g on M with finitely β-class geodesic strings we have:

$$\sum_{o \in \mathcal{O}(g,\beta)} \frac{(-1)^{\mathrm{morse}(o)}}{\mathrm{mult}(o)} = 0.$$

Then the conclusion of the Theorem follows by the same arithmetic argument as in the proof of Theorem 1.5 above. □

7. Proof of Theorem 1.24

For simplicity, as the full argument is already lengthy, we prove the theorem under the assumption that g is Riemannian. In the Finsler case we must use a more modern variational setup for the Hilbert manifold of H^1-loops. In the Finsler case the energy functional is generally only C^1, rather than C^2, on the Hilbert loop space. Nevertheless the required Palais–Smale compactness and critical point theory for closed Finsler geodesics are standard; see Caponio–Javaloyes–Masiello [4], Mercuri [12], and Lu [11]. With these replacements, the compactness argument for end-incompressible classes, and the Fuller-index invariance argument go through without substantial change.

The circle S^1 acts on $L_\beta X$ by reparametrization:

$$(s \cdot \gamma)(t) = \gamma(t+s).$$

If some $\gamma \in L_\beta X$ had nontrivial stabilizer, then γ would factor through a nontrivial covering $S^1 \to S^1$, contradicting indivisibility. Therefore the S^1-action on $L_\beta X$ is free. Hence

$$H_*^{S^1}(L_\beta X, \mathbb{Z}) \cong H_*(L_\beta X/S^1, \mathbb{Z}),$$

and we will construct a finite total rank model for the latter.

Let

$$\mathcal{L}_\beta X = W^{1,2}(S^1, X)_\beta$$

be the Hilbert manifold of H^1-loops in the free homotopy class β. Consider the energy functional

$$\mathcal{E}_g : \mathcal{L}_\beta X \to \mathbb{R}, \qquad \mathcal{E}_g(\gamma) = \int_{S^1} |\dot\gamma(t)|_g^2 \, dt.$$

Since $S(g,\beta)$ is compact, $\mathcal{E}_g$ is bounded above on the critical set of $\mathcal{E}_g$ in class β. Choose

$$E_0 > \max_{\gamma \in S(g,\beta)} \mathcal{E}_g(\gamma) = M^2,$$

which exists by the β-tautness assumption. Set

$$\mathcal{U}_{E_0} = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_g(\gamma) < E_0\}.$$

Lemma 7.1. *The inclusion*

$$\mathcal{U}_{E_0} \hookrightarrow \mathcal{L}_\beta X$$

is an S^1-equivariant homotopy equivalence. Consequently,

$$\mathcal{U}_{E_0}/S^1 \simeq \mathcal{L}_\beta X/S^1.$$

Moreover, the inclusion of smooth loops into the Hilbert loop space is an S^1-equivariant homotopy equivalence, and hence

$$\mathcal{L}_\beta X/S^1 \simeq L_\beta X/S^1.$$

Therefore

$$\mathcal{U}_{E_0}/S^1 \simeq L_\beta X/S^1.$$

Proof. The energy functional $\mathcal{E}_g$ is smooth and S^1-invariant. We equip $\mathcal{L}_\beta X$ with the standard S^1-invariant Hilbert metric induced by g. With respect to this metric, the negative gradient flow of $\mathcal{E}_g$ is S^1-equivariant.

We first recall the Palais–Smale compactness needed for the deformation lemma. Let $\{\gamma_j\} \subset \mathcal{L}_\beta X$ be a Palais–Smale sequence for $\mathcal{E}_g$ with

$$\mathcal{E}_g(\gamma_j) \leq A.$$

Then

$$\ell_g(\gamma_j) \leq \mathcal{E}_g(\gamma_j)^{1/2} \leq A^{1/2}.$$

Since $\beta \in \pi_1^{\mathrm{inc}}(X)$, Lemma 2.1 implies that the images $\gamma_j(S^1)$ lie in a fixed compact subset of X. On this compact set, the usual Hilbert-loop compactness argument for the geodesic energy gives a strongly convergent subsequence in $\mathcal{L}_\beta X$. Thus $\mathcal{E}_g$ satisfies the Palais–Smale condition on sublevels.

By the choice of E_0, the functional $\mathcal{E}_g$ has no critical points in

$$\{\mathcal{E}_g \geq E_0\}.$$

Therefore, for every $A > E_0$, Palais' deformation lemma gives an S^1-equivariant homotopy equivalence

$$\{\mathcal{E}_g < E_0\} \hookrightarrow \{\mathcal{E}_g < A\}.$$

Letting $A \to \infty$, or equivalently applying the negative gradient flow on the exhaustion by sublevels, we obtain an S^1-equivariant homotopy equivalence

$$\mathcal{U}_{E_0} \simeq_{S^1} \mathcal{L}_\beta X.$$

Passing to quotients gives

$$\mathcal{U}_{E_0}/S^1 \simeq \mathcal{L}_\beta X/S^1.$$

Finally, the inclusion

$$C^\infty(S^1, X)_\beta \hookrightarrow W^{1,2}(S^1, X)_\beta$$

is an S^1-equivariant homotopy equivalence by standard equivariant smoothing approximation for loops. Since $L_\beta X$ denotes the same free homotopy component in the smooth or compact-open loop space, we get

$$\mathcal{L}_\beta X \simeq_{S^1} L_\beta X.$$

Combining the quotient equivalences gives

$$\mathcal{U}_{E_0}/S^1 \simeq L_\beta X/S^1.$$

□

We now compare the Fuller index with the equivariant Morse model. The only subtlety is that after perturbing g to a regular metric g', the full class-β geodesic set of g' need not be compact. We therefore work in an isolating neighborhood of the compact orbit block coming from g.

Let g_S^g denote the Sasaki metric on the fixed contact manifold C induced by g under the convention of Section 4. Define the S^1-invariant loop set

$$\widehat{U}_{\sqrt{E_0}} = \{\gamma \in \mathcal{L}_{\widetilde{\beta}} C \mid \ell_{g_S^g}(\gamma) < \sqrt{E_0}\},$$

and let

$$U_{\sqrt{E_0}} = \widehat{U}_{\sqrt{E_0}}/S^1$$

be its image in the orbit-string quotient. By (A.5) and the choice of E_0, this is an isolating neighborhood of

$$N_0 = \mathcal{O}(R^{\lambda_g}, \widetilde{\beta})$$

in the sense of Lemma A.3; it is also controlled for the Sasaki length. Let W be as in Lemma A.3, and let ϵ be the corresponding Fuller constant for the isolating data

$$(N_0, U_{\sqrt{E_0}}, W).$$

Choose a β-regular complete metric g' sufficiently C^2-close to g so that, on the fixed manifold C, the vector field $R^{\lambda_{g'}}$ is $C^0\epsilon$-close to R^{λ_g}. Choose E' with

$$\max_{\gamma\in N_0} \ell_{g_S^g}(\gamma) < \sqrt{E'} < \sqrt{E_0}.$$

Set

$$\widehat{U} = \{\gamma \in \mathcal{L}_{\widetilde{\beta}}C \mid \ell_{g_S^{g'}}(\gamma) < \sqrt{E'}\}, \qquad U = \widehat{U}/S^1.$$

After shrinking the perturbation if necessary, we may assume

$$W \subset U \subset U_{\sqrt{E_0}}.$$

Thus Lemma A.4 gives

$$F(g,\beta) = i(N_0, R^{\lambda_g}, \widetilde{\beta}) = i(N_1, R^{\lambda_{g'}}, \widetilde{\beta}),$$

where

$$N_1 = U \cap \mathcal{O}(R^{\lambda_{g'}}, \widetilde{\beta}).$$

It remains to compute the last Fuller index. Let

$$\mathcal{U}^{g'}_{E'} = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_{g'}(\gamma) < E'\}.$$

For a constant-speed g'-geodesic, its canonical lift has $g_S^{g'}$-length equal to its g'-length. Hence the critical circles of $\mathcal{E}_{g'}$ in $\mathcal{U}^{g'}_{E'}$ are in one-to-one correspondence with the elements of N_1. Since g' is β-regular, this critical set is a finite union of nondegenerate critical circles

$$C_o \simeq S^1.$$

The Morse–Bott handlebody $\mathcal{MB}$ obtained from the unstable disk bundles of these critical circles is S^1-invariant, and the standard Hilbert-manifold Morse–Bott deformation theorem gives an S^1-equivariant homotopy equivalence

$$\mathcal{MB} \simeq_{S^1} \mathcal{U}^{g'}_{E'}.$$

After passing to the quotient by S^1, the negative disk bundle over C_o contributes one cell of dimension $\mathrm{morse}(o)$. Therefore

$$\chi(\mathcal{U}^{g'}_{E'}/S^1) = \chi(\mathcal{MB}/S^1) = \sum_{o\in N_1} (-1)^{\mathrm{morse}(o)}.$$

We now compare the g'-sublevel with the Hilbert sublevel for g. Choose numbers a, b and $\delta > 0$ such that

$$M^2 < a < a+\delta < E' < b < E_0$$

and such that $\mathcal{E}_g$ has no critical points on

$$\{a \le \mathcal{E}_g \le b\}.$$

For g' sufficiently close to g, the sublevel $\mathcal{U}^{g'}_{E'}$ is sandwiched as

$$\mathcal{U}^g_{a+\delta} \subset \mathcal{U}^{g'}_{E'} \subset \mathcal{U}^g_b.$$

By Lemma 7.2, applied with $a+\delta$ in place of a, the inclusion

$$\mathcal{U}^{g'}_{E'} \hookrightarrow \mathcal{U}^g_b$$

is an S^1-equivariant homotopy equivalence. Since all critical points of $\mathcal{E}_g$ have energy at most $M^2 < b$, the inclusion

$$\mathcal{U}^g_b \hookrightarrow \mathcal{U}^g_{E_0}$$

is also an S^1-equivariant homotopy equivalence. Thus

$$\mathcal{U}^{g'}_{E'} \simeq_{S^1} \mathcal{U}^g_{E_0}.$$

By Lemma 7.1,

$$\mathcal{U}^g_{E_0}/S^1 \simeq L_\beta X/S^1.$$

Consequently

$$\chi(\mathcal{U}^{g'}_{E'}/S^1) = \chi(L_\beta X/S^1).$$

In particular $H^{S^1}_*(L_\beta X, \mathbb{Z})$ has finite total rank.

On the other hand, since g' is β-regular, the Fuller index formula gives

$$i(N_1, R^{\lambda_{g'}}, \widetilde{\beta}) = \sum_{o \in N_1} \frac{(-1)^{\mathrm{morse}(o)}}{\mathrm{mult}(o)}.$$

Since β is indivisible, all multiplicities in this class are 1. Hence

$$i(N_1, R^{\lambda_{g'}}, \widetilde{\beta}) = \chi(\mathcal{U}^{g'}_{E'}/S^1) = \chi(L_\beta X/S^1).$$

Therefore

$$F(g, \beta) = \chi(L_\beta X/S^1) = \chi^{S^1}(L_\beta X).$$

□

We state the following elementary lemma separately, since it may be of independent interest.

Lemma 7.2. *For a metric h and a number r, set*

$$\mathcal{U}^h_r = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_h(\gamma) < r\}.$$

Let $a < b$, and suppose that $\mathcal{E}_g$ satisfies the Palais–Smale condition on the band

$$\mathcal{B} = \{a \le \mathcal{E}_g \le b\}$$

and has no critical points there. Let g' be sufficiently C^2-close to g, so that $\nabla\mathcal{E}_{g'}$ is C^0-close to $\nabla\mathcal{E}_g$ on $\mathcal{B}$. Suppose

$$\mathcal{U}^g_a \subset \mathcal{C} \subset \mathcal{U}^g_b,$$

where

$$\mathcal{C} = \{\mathcal{E}_{g'} < c\}$$

is an S^1-invariant sublevel set of $\mathcal{E}_{g'}$. Then the inclusion

$$\mathcal{C} \hookrightarrow \mathcal{U}^g_b$$

is an S^1-equivariant homotopy equivalence.

Proof. By the Palais–Smale condition and the absence of critical points on $\mathcal{B}$, there exists $\mu > 0$ such that

$$\|\nabla\mathcal{E}_g\| \ge \mu \tag{7.3}$$

on $\mathcal{B}$. For g' sufficiently close to g, we have

$$\|\nabla\mathcal{E}_{g'} - \nabla\mathcal{E}_g\| < \mu/2 \tag{7.4}$$

on the same band. Therefore

$$\langle \nabla\mathcal{E}_g, \nabla\mathcal{E}_{g'} \rangle > 0$$

on $\mathcal{B}$.

Since

$$\mathcal{U}^g_a \subset \mathcal{C} \subset \mathcal{U}^g_b,$$

we have

$$\mathcal{U}^g_b - \mathcal{C} \subset \mathcal{U}^g_b - \mathcal{U}^g_a \subset \mathcal{B}.$$

Thus, as long as a negative $\mathcal{E}_{g'}$-gradient trajectory lies in $\mathcal{U}^g_b - \mathcal{C}$, we have

$$\frac{d}{dt}\mathcal{E}_g = -\langle \nabla\mathcal{E}_g, \nabla\mathcal{E}_{g'} \rangle < 0.$$

Consequently the flow cannot exit $\mathcal{U}^g_b$ before entering $\mathcal{C}$. Moreover, $\mathcal{C}$ is forward-invariant because it is a sublevel set of $\mathcal{E}_{g'}$.

The estimates above also imply that $\mathcal{E}_{g'}$ has no critical points on $\mathcal{U}_b^g - \mathcal{C}$. The standard deformation argument of Palais [14, Section 10, Proposition (2)], applied to $\mathcal{E}_{g'}$ on the region $\mathcal{U}_b^g - \mathcal{C}$, therefore gives a deformation retraction of $\mathcal{U}_b^g$ onto $\mathcal{C}$. The energies and the Hilbert metrics are S^1-invariant, so the gradient flow is S^1-equivariant. Hence the deformation retraction is S^1-equivariant. □

8. Proof of Theorem 1.32 and its corollaries

We first prove the following.

Theorem 8.1 (Product formula)**.** *Let*

$$Z \hookrightarrow (X, g) \xrightarrow{p} (Y, g_Y)$$

be a β-periodic fibration, with Y closed, connected and such that all smooth closed contractible g_Y-geodesics in Y are constant.

Fix $y_0 \in Y$, set $Z_0 = p^{-1}(y_0)$, and choose

$$\beta_Z \in \pi_1(Z_0)$$

such that

$$(i_{y_0})_*(\beta_Z) = \beta,$$

where $i_{y_0} : Z_0 \hookrightarrow X$ is the inclusion. Denote by g_{y_0} the restriction of g to the fiber of y_0. Let

$$H = \pi_1(Y, y_0)$$

act on $\pi_1^{\mathrm{inc}}(Z_{y_0})$ by holonomy, and set

$$S_{y_0} = H \cdot \beta_Z.$$

Then

$$F(g, \beta) = (-1)^{\dim Y} \cdot \#S_{y_0} \cdot \chi(Y) \cdot F(g_{y_0}, \beta_Z).$$

Proof. We first show that every class-β g-geodesic in X is vertical.

Let

$$o : S^1 \to X$$

be a closed g-geodesic representing the class β. Since β is a fiber class, the loop

$$p \circ o : S^1 \to Y$$

is contractible. By the definition of a β-periodic fibration, p is a Riemannian submersion and so projects g-geodesics to g_Y-geodesics. Hence $p \circ o$ is a smooth closed contractible g_Y-geodesic. By hypothesis, every such geodesic is constant. Therefore $p \circ o$ is constant, and so o is contained in a single fiber $Z_y = p^{-1}(y)$. Thus the class-β geodesic strings in X are precisely vertical geodesic strings.

Let

$$C = S^*X$$

be the unit cotangent bundle of (X, g), with projection

$$\mathrm{pr} : C \to X.$$

Let R^{λ_g} denote the Reeb vector field on C. Let $\widetilde{\beta}$ be the free homotopy class in C given by the canonical lift of β. We want to compute:

$$F(g, \beta) = i\big(\mathcal{O}(R^{\lambda_g}, \widetilde{\beta}), R^{\lambda_g}, \widetilde{\beta}\big).$$

The idea is to localize contributions over critical points of a Morse function on Y.

Let

$$\pi^{\mathrm{vert}} : TX \to T^{\mathrm{vert}}X = \ker(dp)$$

be the g-orthogonal projection. By assumptions on $p : X \to Y$,

$$\nabla \pi^{\mathrm{vert}} = 0,$$

where π^{vert} is viewed as a $(1,1)$-tensor field.

For $\xi \in C$, let

$$v_\xi \in T_{\mathrm{pr}(\xi)} X$$

be the g-dual unit vector to ξ. Define

$$P : C \to \mathbb{R}$$

by

$$P(\xi) = |\pi^{\mathrm{vert}} v_\xi|_g^2.$$

Thus $P = 1$ on vertical unit covectors and $P = 0$ on horizontal unit covectors.

Let $f : Y \to \mathbb{R}$ be a Morse function. Define

$$\widetilde{f} : C \to \mathbb{R}$$

by

$$\widetilde{f} = f \circ p \circ \mathrm{pr}\,.$$

Equip $C = S^*X$ with the Sasaki metric g_S induced by g, as in the Proof of Theorem 1.24. We shall use the following two elementary identities.

First,

$$dP(R^{\lambda_g}) = 0. \tag{8.2}$$

Indeed, let $\xi(\tau)$ be a flow line of R^{λ_g}, and set

$$x(\tau) = \mathrm{pr}(\xi(\tau)).$$

Then $x(\tau)$ is a unit-speed g-geodesic. Let

$$v(\tau) = \dot{x}(\tau) = v_{\xi(\tau)}$$

be its velocity vector field. Since $x(\tau)$ is a geodesic,

$$\nabla_\tau v = 0.$$

Since $\nabla \pi^{\mathrm{vert}} = 0$, we have

$$\nabla_\tau(\pi^{\mathrm{vert}} v) = \pi^{\mathrm{vert}}(\nabla_\tau v) = 0.$$

Therefore

$$\begin{aligned} \frac{d}{d\tau} P(\xi(\tau)) &= \frac{d}{d\tau} |\pi^{\mathrm{vert}} v(\tau)|_g^2 \\ &= 2 \left\langle \nabla_\tau(\pi^{\mathrm{vert}} v), \pi^{\mathrm{vert}} v \right\rangle_g \\ &= 0. \end{aligned}$$

Hence $dP(R^{\lambda_g}) = 0$.

Second,

$$dP(\nabla_{g_S} \widetilde{f}) = 0. \tag{8.3}$$

Since $\widetilde{f} = f \circ p \circ \mathrm{pr}$ depends only on the base point $\mathrm{pr}(\xi)$:

$$\nabla_{g_S} \widetilde{f} \in T^{\mathrm{hor}} C.$$

More precisely, $\nabla_{g_S} \widetilde{f}$ is the horizontal lift of

$$\nabla_g (f \circ p)$$

on X. Let $\xi(s)$ be a horizontal curve in C, and set

$$x(s) = \mathrm{pr}(\xi(s)).$$

By definition of the Levi-Civita horizontal distribution on S^*X, the covectors $\xi(s)$ are parallel along $x(s)$. Equivalently, their g-dual vectors

$$v(s) = v_{\xi(s)}$$

are parallel along $x(s)$:

$$\nabla_s v(s) = 0.$$

Using again $\nabla \pi^{\mathrm{vert}} = 0$, we get

$$\nabla_s(\pi^{\mathrm{vert}} v(s)) = \pi^{\mathrm{vert}}(\nabla_s v(s)) = 0.$$

Hence

$$\begin{aligned}\frac{d}{ds}P(\xi(s)) &= \frac{d}{ds}|\pi^{\mathrm{vert}} v(s)|_g^2 \\ &= 2\left\langle \nabla_s(\pi^{\mathrm{vert}} v(s)), \pi^{\mathrm{vert}} v(s)\right\rangle_g \\ &= 0.\end{aligned}$$

Taking $\dot{\xi}(0) = \nabla_{g_S} \widetilde{f}(\xi)$ gives

$$dP_\xi(\nabla_{g_S}\widetilde{f}) = 0.$$

Since ξ was arbitrary,

$$dP(\nabla_{g_S}\widetilde{f}) = 0.$$

Now set

$$V_\varepsilon = R^{\lambda_g} - \varepsilon \nabla_{g_S}(P + \widetilde{f}).$$

Recall first the discussion of Fuller continuation in the Appendix.

$$N_0 = \mathcal{O}(R^{\lambda_g}, \widetilde{\beta})$$

is compact by β-tautness, take $\varepsilon > 0$ to be small enough so that V_ϵ is C^0 ϵ'-close to R^{λ_g}, where ϵ' is the Fuller constant of N_0. The Fuller continuation of N_0 is a compact set $N_\varepsilon \subset \mathcal{O}(V_\epsilon, \widetilde{\beta})$ and

$$i\big(N_0, R^{\lambda_g}, \widetilde{\beta}\big) = i(N_\varepsilon, V_\varepsilon, \widetilde{\beta}).$$

Note that the above does not require any further properties of V_ϵ, like completeness. We setup the Fuller index theory to sidesteps such issues.

We now identify the closed orbits in N_ε. We show that P is a Lyapunov function for V_ϵ. Mainly, along a trajectory $\xi(\tau)$ of V_ε, we have

$$\begin{aligned}\frac{d}{d\tau}P(\xi(\tau)) &= dP(R^{\lambda_g}) - \varepsilon dP(\nabla_{g_S}P) - \varepsilon dP(\nabla_{g_S}\widetilde{f}) \\ &= -\varepsilon|\nabla_{g_S}P|_{g_S}^2 \leq 0.\end{aligned}$$

Thus P is nonincreasing along the V_ε-flow.

If $\xi(\tau)$ is a closed orbit of V_ε, then $P(\xi(\tau))$ returns to its initial value after one period. Since P is nonincreasing, it must be constant along the orbit. Hence

$$\nabla_{g_S}P = 0$$

along the orbit.

On each unit cotangent sphere, P is the squared norm of the vertical component of the corresponding unit vector. Therefore the critical locus of P consists of the two critical loci

$$P = 1 \qquad \text{and} \qquad P = 0,$$

corresponding respectively to vertical and horizontal covectors. At $\varepsilon = 0$, the class-$\widetilde{\beta}$ orbit set under consideration is vertical, hence lies in $P = 1$. For $\varepsilon > 0$ sufficiently small, the continued set N_ε remains in the continuation of this vertical component. Thus $P = 0$ is excluded, and every closed orbit in N_ε lies in the vertical critical locus $P = 1$.

Consequently, every closed orbit in N_ε projects to a geodesic contained in a fiber Z_y.

It remains to see over which fibers these vertical orbits occur. On $P = 1$, the Reeb vector field restricts to the geodesic Reeb field of the fiber (Z_y, g_y), while $\widetilde{f}$ depends only on the base point y. The perturbation

$$-\varepsilon \nabla_{g_S} \widetilde{f}$$

moves the base point in the negative gradient direction of f. Therefore a vertical closed orbit of V_ε can occur only over a critical point of f. Conversely, if $y \in \mathrm{Crit}(f)$, then

$$\nabla_{g_S} \widetilde{f} = 0$$

over Z_y, and the vertical part of V_ε is just the fiber geodesic flow, up to the harmless P-term which vanishes on the vertical critical locus. Thus the closed orbits of V_ε in N_ε are precisely the vertical closed geodesic orbits lying in fibers over critical points of f.

We now compute the index. For $y \in Y$, put

$$H_y = \pi_1(Y, y),$$

acting on $\pi_1^{\mathrm{inc}}(Z_y)$ by holonomy. Choose

$$\beta_Z \in \pi_1^{\mathrm{inc}}(Z_y)$$

with

$$(i_y)_*(\beta_Z) = \beta.$$

Set

$$S_y = H_y \cdot \beta_Z.$$

Now the Fuller index is additive, that is N, N' are compact open and $N \cap N' = \emptyset$ then

$$i(N + N', V, \beta) = i(N, V, \beta) + i(N', V, \beta).$$

So it remain to compute the local index contribution $i(N_{\epsilon,y}, V_\epsilon, \widetilde{\beta})$ corresponding to the isolated subset of N_ϵ lying over a critical point y of f. We defer this computation to the Appendix as Lemma B.1; it yields that:

$$i(N_{\varepsilon,y}, V_\varepsilon, \widetilde{\beta}) = (-1)^{\dim Y + \mathrm{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha), \tag{8.4}$$

Next, we show that $F(g_y, \cdot)$ is constant on S_y. Indeed, let

$$\gamma : [0, 1] \to Y$$

be a path based at y. Pulling back $p : X \to Y$ along γ gives a fibration over $[0, 1]$. By the fiber-tautness condition in the definition of a β-periodic fibration, this pullback is a taut deformation of the fiber metric from (Z_y, g_y) back to itself, with final fiber identified with the initial one by the holonomy map $h_\gamma : Z_y \to Z_y$. Fuller index invariance under taut deformation gives

$$F(g_y, \alpha) = F(g_y, h_{\gamma,*}\alpha).$$

Thus $F(g_y, \alpha)$ is constant on the holonomy orbit S_y. Hence

$$\sum_{\alpha \in S_y} F(g_y, \alpha) = \#S_y \cdot F(g_y, \beta_Z).$$

If $y, y' \in Y$, holonomy along any path from y to y' identifies S_y with $S_{y'}$. The same path gives a taut deformation between the fiber metrics, so

$$\sum_{\alpha \in S_y} F(g_y, \alpha)$$

is independent of y. So we get:

$$\begin{aligned}
F(g,\beta) &= \sum_{y\in \mathrm{Crit}(f)} i(N_{\varepsilon,y}, V_\varepsilon, \widetilde{\beta}) \\
&= \sum_{y\in \mathrm{Crit}(f)} (-1)^{\dim Y + \mathrm{morse}(y)} \sum_{\alpha\in S_y} F(g_y,\alpha), \\
&= \sum_{y\in \mathrm{Crit}(f)} (-1)^{\dim Y + \mathrm{morse}(y)} \cdot \# S_{y_0} \cdot F(g_{y_0}, \beta_Z) \\
&= (-1)^{\dim Y} \cdot \chi(Y) \cdot \# S_{y_0} \cdot F(g_{y_0}, \beta_Z).
\end{aligned}$$

□

We return to the proof of Theorem 1.32. By Cartan–Hadamard, a complete negatively curved Riemannian manifold has at most one closed geodesic in each nontrivial free homotopy class; moreover the absence of nontrivial periodic Jacobi fields in negative curvature gives nondegeneracy. Hence, the first part follows from the second.

We prove second part. Suppose first that β is an n-power, so that β has a representative with multiplicity n, and we let α denote the corresponding indivisible class. By the assumption that all contractible g_Y geodesics are constant, classical Morse theory Milnor [13] tells us that Y has vanishing higher homotopy groups $\pi_k(Y, y_0)$, $k \geq 2$. Thus by the long exact sequence of a fibration $i_{Z,*} : \pi_1(Z, z_0) \to \pi_1(X, i_Z(z_0))$ is a group injection, and image $i_{Z,*} = \ker p_*$. Now $\pi_1(Y, y_0)$ is torsion-free since it is a finite dimensional $K(\pi_1(Y, y_0), 1)$ space, Hatcher [9, Prop. 2.45]. It follows that α is also a fiber class since

$$p_*(\alpha^n) = 0 \implies p_*(\alpha) = 0.$$

Furthermore, $\alpha \in \pi_1^{inc}(X)$ (otherwise β isn't).

Lemma 8.5 (Power propagation of periodicity)**.** *Let $p : Z \hookrightarrow (X, g) \to (Y, g_Y)$ be a g-parallel Riemannian submersion with complete total space, taut fibers, and totally geodesic fibers (automatic from $\nabla\pi^{\mathrm{vert}} = 0$). Let $\beta \in \pi_1^{\mathrm{inc}}(X)$ be a fiber class which is β-taut, and suppose $\beta = \gamma^m$ for some $\gamma \in \pi_1^{\mathrm{inc}}(X)$, $m \geq 1$. If p is β-holonomy-finite, then p is also γ-holonomy-finite and γ-taut; in particular p is a γ-periodic fibration.*

Proof. Conditions (1), (2), (4) of Definition 1.31 are independent of the class, so we need only establish γ-tautness and γ-holonomy-finiteness.

γ-tautness. Every closed γ-geodesic o_γ has m-fold cover $o_\gamma^m \in S(g, \beta)$, so

$$\ell_g(o_\gamma) = \tfrac{1}{m}\, \ell_g(o_\gamma^m) \leq A/m, \qquad A := \max_{o\in S(g,\beta)} \ell_g(o) < \infty,$$

where finiteness of A is β-tautness. Hence $S(g, \gamma) = S_{A/m}(g, \gamma)$, which is compact by Lemma 2.2. This step uses only β-tautness and is independent of any holonomy condition.

γ-holonomy-finiteness. Lift γ to $\gamma_Z \in \pi_1(Z_{y_0})$ and set $S_{y_0} = \pi_1(Y, y_0) \cdot \gamma_Z$. Suppose S_{y_0} were infinite. Holonomy preserves π_1^{inc}, so each class in S_{y_0} lies in $\pi_1^{\mathrm{inc}}(Z_{y_0})$; as the fiber metric is complete, each carries a closed geodesic by Lemma 2.3. These geodesics are vertical, hence pairwise distinct elements of $S(g, \gamma)$. By Lemma 2.1 they are confined to a fixed compact set with lengths $\leq A/m$, so by Arzelà–Ascoli a subsequence converges in C^0 to a closed geodesic o'_∞. Since p is g-parallel its fibers are totally geodesic; thus for large k the geodesic homotopy from o'_{i_k} to o'_∞ is vertical and gives a free homotopy *within* Z_{y_0}. Hence the corresponding classes eventually coincide, contradicting their distinctness. Therefore S_{y_0} is finite and p is γ-holonomy-finite.

Together with γ-tautness, p is a γ-periodic fibration. □

By the lemma above p is an α-periodic fibration. Now, if $\chi(Y) \neq \pm 1$ then by the product formula $F(g, \alpha) \neq \pm 1$, since $F(g_{y_0}, \alpha_Z)$ is an integer by Theorem 1.24.

Now, if X admits a complete metric h with a unique and nondegenerate class-β h-geodesic string o_β, then it clearly also has a unique class-α h-geodesic string o_α. Furthermore, o_α must be non-degenerate as otherwise o_β, which is a k-fold cover of o_α is degenerate. Then we have:

$$F(h, \alpha) = \pm 1.$$

But this is a contradiction.

We now prove the general case. Suppose by contradiction that X admits a complete metric h with a unique and nondegenerate class-β h-geodesic string o_β. By the preceding paragraph, the class β is not atomic. In other words, β is not a finite power of a class which is not itself a proper power; equivalently, in the present terminology, β is infinitely divisible. Now o_β covers a multiplicity one geodesic string o_α in some class $\alpha \in \pi_1^{inc}(X)$. Moreover, o_α is the unique geodesic string in its class, otherwise o_β would not be unique in its class. We prove that α is indivisible, which will be a contradiction to β not being atomic and will complete the proof.

Suppose otherwise, so that $\alpha_{x_0} = \rho^k$ for $k > 1$ and $\rho \in \pi_1(X, x_0)$, (with the notation of Definition 1.22). Let o_ρ be a class ρ, closed g-geodesic string (where by slight abuse ρ also denotes the class in $\pi_1^{inc}(X)$ corresponding to the based class ρ.) It is immediate that the k cover of o_ρ, o_ρ^k represents α and is a g-geodesic string. By the uniqueness, $o_\alpha = o_\rho^k$. But this contradicts simplicity of o_α. So α is indivisible. Since o_β is a finite cover of o_α, the class β is a finite power of the indivisible class α. Hence β is atomic, contradicting the conclusion of the first part of the proof.

□

Proof of Theorem 1.6. Let g_Z, g_Y be complete Riemannian metrics on Z with g_Z having non-positive sectional curvature and g_Y not admitting contractible closed geodesics. Take the product metric $g = g_Z \times g_Y$ on $X = Z \times Y$. For a class $\beta \in \pi_1^{inc}(X)$ in the image of the inclusion $\pi_1^{inc}(Z) \to \pi_1^{inc}(X)$, the natural projection $X \to Y$ is automatically a β-periodic fibration. Then the conclusion readily follows from Theorem 1.32.

□

Proof of Theorem 1.26. By Theorem 8.1 0 is certainly a value of the invariant F. We first prove that every negative rational number is the value of the invariant. Let p, q be positive integers. Let Y be a closed surface of genus $(p+1) > 1$ with a hyperbolic metric g_Y, let Z be the genus 2 closed surface with a hyperbolic metric g_Z and let $\beta_Z \in \pi_1^{inc}(Z)$ be the class represented by a $2 \cdot q$-fold covering of a simple closed loop representing a generator of the fundamental group of Z.

Let $X = Y \times Z$ with the product metric $g = g_Y \times g_Z$ and $p : X \to Y$ the canonical projection. By Theorem 8.1 since $(-1)^{\dim Y} = 1$ and $\# S_{y_0} = 1$,

$$F(g, \beta) = \chi(Y) \cdot F(g_Z, \beta_Z) = (-2p) \cdot \frac{1}{2q} = -\frac{p}{q},$$

where $\beta = (i_Z)_*(\beta_Z)$. So we proved our first claim.

Let again p, q be positive integers. Let Y be closed surface of genus 2, with a hyperbolic metric g_Y. And let Z be a manifold satisfying $F(g_Z, \beta_Z) = -\frac{p}{2q}$ for some β_Z-taut metric g_Z on Z and for some class $\beta_Z \in \pi_1^{inc}(Z)$. This exist by the discussion above. Let $g = g_Y \times g_Z$ be the product metric on $Y \times Z$, and β as above. Analogously to the discussion above we get:

$$F(g, \beta) = \chi(Y) \cdot F(g_Z, \beta_Z) = (-2) \cdot \frac{-p}{2q} = \frac{p}{q}.$$

□

Appendix A. Fuller index and sky catastrophes

In this appendix we recall the part of Fuller's periodic orbit index used in the paper. We also fix the notation for orbit strings and for continuation under small perturbations of vector fields.

Let X be a smooth manifold, and fix a complete auxiliary Riemannian metric h on X. Let

$$\mathcal{V}(X)$$

denote the space of complete smooth nonsingular vector fields on X, with the C^0-topology on compact subsets.

For $V \in \mathcal{V}(X)$ and a free homotopy class β, define

$$S(V, \beta) = \{o \in L_\beta X \mid \dot{o}(t) = T\, V(o(t)) \text{ for some } T > 0\}\,.$$

The number T is uniquely determined by o, since V has no zeros. We call it the period of o, and write

$$T = T(o).$$

The circle $S^1 = \mathbb{R}/\mathbb{Z}$ acts on $S(V, \beta)$ by reparametrization:

$$(s \cdot o)(t) = o(t + s).$$

We define the space of orbit strings by

$$\mathcal{O}(V, \beta) = S(V, \beta)/S^1.$$

When no confusion is possible, we use the same letter o for a parametrized orbit and for its associated orbit string.

Let

$$q : L_\beta X \to L_\beta X/S^1$$

be the quotient map. If

$$U \subset L_\beta X/S^1,$$

we write

$$\widehat{U} = q^{-1}(U) \subset L_\beta X.$$

We say that U is h-length bounded if the h-lengths of all loops in $\widehat{U}$ are uniformly bounded.

For an orbit string $o \in \mathcal{O}(V, \beta)$, its multiplicity

$$m(o) \in \mathbb{N}$$

is the unique positive integer such that o is the $m(o)$-fold cover of a primitive orbit string. Equivalently, if $T(o)$ is the period of o and $T_{\text{prim}}(o)$ is the period of the underlying primitive orbit string, then

$$m(o) = \frac{T(o)}{T_{\text{prim}}(o)}.$$

A.1. **Fuller index.** We first recall the definition in the nondegenerate case. Suppose

$$N \subset \mathcal{O}(V, \beta)$$

is finite and consists of nondegenerate orbit strings. Fuller associates to (N, V, β) the rational number

$$i(N, V, \beta) = \sum_{o \in N} \frac{i(o)}{m(o)}.$$

Here $i(o)$ is the fixed point index of the time-$T(o)$ return map of the flow of V, computed on a local hypersurface transverse to the image of o.

By a compact open subset of an orbit-string space, we mean a subset which is open in the relative topology and compact as a subset. For a general compact open subset

$$N \subset \mathcal{O}(V, \beta),$$

the Fuller index can be defined by first making a sufficiently small perturbation of V, supported in an isolating neighborhood of N, so that the corresponding orbit strings become nondegenerate. Fuller's construction shows that the resulting number is independent of the auxiliary perturbation.

In the special case of a nondegenerate closed geodesic orbit, the local fixed point index agrees with the usual Morse sign:

$$i(o) = (-1)^{\mathrm{morse}(o)}.$$

Consequently, for a regular geodesic flow, Fuller's index becomes

$$i(N, V, \beta) = \sum_{o \in N} \frac{(-1)^{\mathrm{morse}(o)}}{m(o)}.$$

A.2. **Continuation.** Let

$$\{V_t\}_{t \in [0,1]} \subset \mathcal{V}(X)$$

be a continuous family. Define

$$S(\{V_t\}, \beta) = \{(o, t) \in L_\beta X \times [0,1] \mid o \in S(V_t, \beta)\},$$

and

$$\mathcal{O}(\{V_t\}, \beta) = S(\{V_t\}, \beta)/S^1, \tag{A.1}$$

where S^1 acts on the loop coordinate.

Fuller's basic invariance says the following. Let

$$\mathcal{N} \subset \mathcal{O}(\{V_t\}, \beta)$$

be compact and open. Suppose that its endpoint slices are

$$N_0 = \mathcal{N} \cap \mathcal{O}(V_0, \beta), \qquad N_1 = \mathcal{N} \cap \mathcal{O}(V_1, \beta).$$

Then

$$i(N_0, V_0, \beta) = i(N_1, V_1, \beta). \tag{A.2}$$

We refer to this as the basic invariance of the Fuller index.

For applications, we need a local version of this statement under small perturbations. This is a variation of [8, Lemma 4.1], with the added complexity needed for dealing with noncompact X.

Lemma A.3 (Isolation lemma)**.** *Let h be a complete auxiliary Riemannian metric on X. Let V_0 be a smooth nonsingular vector field on X, and let*

$$N_0 \subset \mathcal{O}(V_0, \beta)$$

be compact and open. Let

$$U \subset L_\beta X/S^1$$

be an open neighborhood of N_0 such that

$$\overline{U} \cap \mathcal{O}(V_0, \beta) = N_0.$$

Assume that U is controlled: there are a compact set $K \subset X$ and a constant $L > 0$ such that every loop $o \in \widehat{U}$ satisfies

$$\mathrm{image}(o) \subset K, \qquad \ell_h(o) \le L.$$

Then for every open neighborhood W of N_0 satisfying

$$\overline{W} \subset U,$$

there exists $\epsilon > 0$ with the following property. If V is a smooth vector field such that

$$\sup_{x \in K} |V(x) - V_0(x)|_h < \epsilon,$$

then

$$\mathcal{O}(V, \beta) \cap U \subset W.$$

In particular,

$$\mathcal{O}(V, \beta) \cap U$$

is compact open.

Proof. Since V_0 is nonsingular and K is compact, there are constants

$$0 < B < A < \infty$$

such that

$$B \leq |V_0(x)|_h \leq A$$

for all $x \in K$. We shall choose $\epsilon < B/2$. Then every vector field V satisfying

$$\sup_K |V - V_0|_h < \epsilon$$

obeys

$$|V(x)|_h \geq B/2, \qquad |V(x)|_h \leq A + \epsilon$$

for all $x \in K$.

Suppose the conclusion failed for the chosen W. Then there would be a sequence of smooth vector fields

$$V_j \to V_0 \qquad \text{uniformly on } K,$$

and orbit strings

$$o_j \in \big(\mathcal{O}(V_j, \beta) \cap U\big) - W.$$

Choose parametrized representatives, still denoted o_j, satisfying

$$\dot{o}_j(s) = T_j V_j(o_j(s)).$$

Since $o_j \in \widehat{U}$, controlledness gives

$$\operatorname{image}(o_j) \subset K, \qquad \ell_h(o_j) \leq L.$$

For j sufficiently large, $|V_j|_h \geq B/2$ on K. Hence

$$\ell_h(o_j) = T_j \int_0^1 |V_j(o_j(s))|_h \, ds \geq T_j B/2.$$

Thus

$$T_j \leq \frac{2L}{B}.$$

Similarly, for j sufficiently large,

$$|V_j|_h \leq A + \epsilon$$

on K, and therefore

$$|\dot{o}_j(s)|_h = T_j |V_j(o_j(s))|_h \leq \frac{2L(A+\epsilon)}{B}.$$

Thus the o_j are equicontinuous and have images in the fixed compact set K. By Arzelà–Ascoli, after passing to a subsequence,

$$o_j \to o_\infty$$

uniformly. Also, after passing to a further subsequence, the periods converge:

$$T_j \to T_\infty.$$

Since $V_j \to V_0$ uniformly on K, passing to the limit in

$$\dot{o}_j = T_j V_j(o_j)$$

shows that

$$\dot{o}_\infty = T_\infty V_0(o_\infty).$$

Moreover $T_\infty > 0$, since

$$\ell_h(o_j) \geq T_j B/2$$

and the orbits are nonconstant in the class β. Hence o_∞ is a V_0-orbit in class β.

Now $o_\infty \in \overline{U} - W$. This contradicts

$$\overline{U} \cap \mathcal{O}(V_0, \beta) = N_0 \subset W.$$

Therefore

$$\mathcal{O}(V, \beta) \cap U \subset W$$

for all V sufficiently C^0-close to V_0 on K.

Note that $\mathcal{O}(V, \beta) \cap U = \mathcal{O}(V, \beta) \cap \overline{W}$, hence $\mathcal{O}(V, \beta) \cap U$ is closed. The compactness of $\mathcal{O}(V, \beta) \cap U$ then follows from the same argument. Every sequence of orbit strings in this set has representatives with image in K, length bounded by L, period bounded by $2L/B$, and uniformly bounded velocity. Thus Arzelà–Ascoli gives a convergent subsequence, and the limit is again an orbit string lying in $\mathcal{O}(V, \beta) \cap U$. □

We will call ϵ above the ***Fuller constant*** of V_0, N_0, U, W (a priori it also depends on K, L). If we just say Fuller constant this data is implicitly chosen.

Lemma A.4 (Fuller continuation)**.** *Let V_0, N_0, U, W, K be as in Lemma A.3, and let ϵ be the associated Fuller constant. Let V_1 be ϵ C^0-close to V_0 on K, and set*

$$V_t = (1-t)V_0 + tV_1.$$

Then

$$\mathcal{N} = (U \times [0,1]) \cap \mathcal{O}(\{V_t\}, \beta)$$

is compact and open in $\mathcal{O}(\{V_t\}, \beta)$. Its endpoint slices are

$$N_0 = U \cap \mathcal{O}(V_0, \beta), \qquad N_1 = U \cap \mathcal{O}(V_1, \beta).$$

Consequently,

$$i(N_0, V_0, \beta) = i(N_1, V_1, \beta).$$

We call N_1 the **Fuller continuation** *of N_0.*

Proof. $(U \times [0,1]) \cap \mathcal{O}(\{V_t\}, \beta)$ is clearly open, it is also closed since by Lemma A.3

$$(U \times [0,1]) \cap \mathcal{O}(\{V_t\}, \beta) = (\overline{W} \times [0,1]) \cap \mathcal{O}(\{V_t\}, \beta).$$

The compactness follows by the same argument as in Lemma A.3. The endpoint slices are plainly

$$\mathcal{N} \cap \mathcal{O}(V_0, \beta) = U \cap \mathcal{O}(V_0, \beta) = N_0$$

and

$$\mathcal{N} \cap \mathcal{O}(V_1, \beta) = U \cap \mathcal{O}(V_1, \beta) = N_1.$$

Fuller's basic invariance applied to the compact open cobordism $\mathcal{N}$ gives

$$i(N_0, V_0, \beta) = i(N_1, V_1, \beta).$$

□

Example 4 (Geodesic flows and the Sasaki metric). Let (Y, g_Y) be a complete Riemannian manifold, and let

$$C_g = S^*Y$$

be its unit cotangent bundle with Liouville contact form λ_{g_Y}. The Reeb vector field $R^{\lambda_{g_Y}}$ is the geodesic Reeb field.

Let g_S be the Sasaki metric on S^*Y. It is defined using the Levi-Civita connection of g_Y. Thus

$$TC = T^{\mathrm{vert}}C \oplus T^{\mathrm{hor}}C$$

is an orthogonal splitting, where

$$T^{\mathrm{vert}}C = \ker(\mathrm{pr}_*)$$

and $T^{\mathrm{hor}}C$ is the Levi-Civita horizontal subbundle. If $\xi, \eta \in T(S^*Y)$ are horizontal, then

$$g_S(\xi, \eta) = g_Y(\mathrm{pr}_* \xi, \mathrm{pr}_* \eta),$$

and the vertical part is measured using the metric induced by g_Y on the cotangent fibers.

The basic property needed in the paper is the following. If o is a nonzero constant-speed g_Y-geodesic and $\widetilde{o}$ is its unit cotangent lift, then

$$\ell_{g_S}(\widetilde{o}) = \ell_{g_Y}(o). \tag{A.5}$$

Indeed, the covector component of $\widetilde{o}$ is parallel along o, so the vertical part of $\dot{\widetilde{o}}$ vanishes and the Sasaki speed equals the base speed.

If $\beta \in \pi_1^{\mathrm{inc}}(Y)$ and $U \subset S(g_Y, \beta)$ is g_Y-length bounded, then the set of canonical lifts

$$\widetilde{U} \subset S(R^{\lambda_{g_Y}}, \widetilde{\beta})$$

is g_S-length bounded by (A.5).

Appendix B. A local fibered calculation of F

We prove in this section the technical lemma computing the local index, used in Theorem 8.1.

Let

$$P : S^*X \to \mathbb{R}$$

be the verticality function as in the proof of Theorem 8.1,

$$P(\xi) = |\pi^{\mathrm{vert}} v_\xi|_g^2.$$

Here v_ξ is the g-dual unit vector to ξ, and

$$\pi^{\mathrm{vert}} : TX \to T^{\mathrm{vert}}X$$

is the g-orthogonal projection. The notation β, $\widetilde{\beta}$, and β_Z is likewise as in Theorem 8.1.

Lemma B.1 (Local product calculation)**.** *Let*

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a parallel Riemannian submersion. Let

$$\widetilde{f} = f \circ p \circ \mathrm{pr}$$

for a Morse function $f : Y \to \mathbb{R}$, and define the vector field on S^*X

$$V_\varepsilon = R^{\lambda_g} - \varepsilon \nabla_{g_S}(P + \widetilde{f}).$$

Fix

$$y \in \mathrm{Crit}(f),$$

and let

$$N_{\varepsilon,y}$$

be the isolated set of closed V_ε-orbit strings lying over the fiber

$$Z_y = p^{-1}(y),$$

as in the proof of Theorem 8.1. Then

$$i(N_{\varepsilon,y}, V_\varepsilon, \widetilde{\beta}) = (-1)^{\dim Y + \mathrm{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha),$$

where S_y is the $\pi_1(Y,y)$-orbit of the corresponding fiber representative in $\pi_1^{\mathrm{inc}}(Z_y)$. In particular, if the fibration is holonomy-finite in the relevant class, the sum is finite.

Proof. Let

$$T^{\mathrm{vert}}X = \ker(dp), \qquad H = (T^{\mathrm{vert}}X)^{\perp_g}.$$

Since the submersion is parallel, the two distributions

$$T^{\mathrm{vert}}X \quad \text{and} \quad H$$

are preserved by the Levi-Civita connection. Since $T^{\mathrm{vert}}X$ and $H = (T^{\mathrm{vert}}X)^{\perp_g}$ are parallel complementary distributions, the local de Rham decomposition theorem gives a product decomposition

$$p^{-1}(D_y) \cong Z_y \times D_y, \qquad g = g_y \oplus g_Y|_{D_y},$$

for a normal ball neighborhood $D_y \subset Y$ centered at y. See Kobayashi–Nomizu [10, Chapter IV].

For this local calculation we suppress the fixed-C identification and write S^*X for the g-unit cotangent bundle. Let

$$C^{\mathrm{vert}} = \{P = 1\} \subset S^*X$$

be the vertical locus. Over Z_y, this locus is naturally identified with

$$S^*Z_y \subset S^*X.$$

Near

$$S^*Z_y \subset C^{\mathrm{vert}},$$

the product decomposition gives the normal-bundle model

$$S^*Z_y \times D_y \times B_y^*, \tag{B.2}$$

where

$$B_y^* \subset T_y^*Y$$

is a small ball in the horizontal cotangent direction. In this model, the vertical locus is

$$S^*Z_y \times D_y \times \{0\}.$$

A covector near the vertical locus decomposes as

$$\xi = \xi_{\mathrm{vert}} + \eta, \qquad \eta \in B_y^*.$$

The unit condition is

$$|\xi_{\mathrm{vert}}|^2_{g_y^*} + |\eta|^2_{g_Y^*} = 1.$$

Therefore

$$P(\xi) = |\xi_{\mathrm{vert}}|^2_{g_y^*} = 1 - |\eta|^2_{g_Y^*}.$$

Thus P has a Morse–Bott maximum along C^{vert}.

In the local model (B.2),

$$N_{\varepsilon,y} = N_y^{\mathrm{fib}} \times \{y\} \times \{0\},$$

where N_y^{fib} is the relevant closed orbit set of the fiber geodesic Reeb flow on S^*Z_y.

We now describe the local return map at an element of $N_{\varepsilon,y}$. Along the vertical locus over Z_y, the fiber component of V_ε is the geodesic Reeb field $R^{\lambda_{g_y}}$ on S^*Z_y. The base component is

$$-\varepsilon \nabla_{g_Y} f$$

on D_y. The normal covector component is the outward radial component coming from

$$-\varepsilon\nabla_{g_S}P$$

on B_y^*.

Let

$$o \in N_y^{\mathrm{fib}},$$

and choose a product transverse section

$$\Sigma = \Sigma^{\mathrm{fib}} \times D_y \times B_y^*,$$

where

$$\Sigma^{\mathrm{fib}} \subset S^*Z_y$$

is transverse to the fiber Reeb flow near o. With respect to this product section, the local return map of V_ε is the product

$$\Phi_{\mathrm{fib}} \times \psi_y \times \kappa.$$

Here:

- Φ_{fib} is the local return map of the fiber geodesic flow generated by $R^{\lambda_{g_y}}$;
- ψ_y is the small-time map of

  $$-\varepsilon\nabla_{g_Y}f$$

  near y;
- κ is the normal return map induced by

  $$-\varepsilon\nabla_{g_S}P$$

  near $0 \in B_y^*$.

The fixed point index is multiplicative under products, Dold [5, Chapter VII, Section 5]. Hence

$$i_{\mathrm{fp}}(o; V_\varepsilon) = i_{\mathrm{fp}}(o; R^{\lambda_{g_y}}) \cdot \mathrm{ind}_{PH}(-\nabla_{g_Y}f, y) \cdot \mathrm{ind}_{PH}(\kappa, 0).$$

Since y is a Morse critical point,

$$\mathrm{ind}_{PH}(-\nabla_{g_Y}f, y) = (-1)^{\mathrm{morse}(y)}.$$

Since the normal component of

$$-\varepsilon\nabla_{g_S}P$$

has outward radial linearization on a vector space of dimension $\dim Y$,

$$\mathrm{ind}_{PH}(\kappa, 0) = (-1)^{\dim Y}.$$

Therefore

$$i_{\mathrm{fp}}(o; V_\varepsilon) = (-1)^{\dim Y + \mathrm{morse}(y)} i_{\mathrm{fp}}(o; R^{\lambda_{g_y}}).$$

The multiplicity of o is unchanged by taking the product with the isolated base and normal fixed points. Thus

$$\frac{i_{\mathrm{fp}}(o; V_\varepsilon)}{m(o)} = (-1)^{\dim Y + \mathrm{morse}(y)} \frac{i_{\mathrm{fp}}(o; R^{\lambda_{g_y}})}{m(o)}.$$

Summing over all fiber orbit strings in the local block gives

$$i(N_{\varepsilon,y}, V_\varepsilon, \widetilde{\beta}) = (-1)^{\dim Y + \mathrm{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha).$$

This proves the claim. $\square$

Appendix C. Sky catastrophes

We now recall the topological formulation of sky catastrophes. This is the version used in Fuller's theory and in the applications above.

Definition C.1. *Let*
$$\{V_t\}_{t\in[0,1]} \subset \mathcal{V}(X)$$
be a continuous family. We say that the family has a sky catastrophe in class β if there is an endpoint orbit string
$$y \in \mathcal{O}(V_0,\beta) \cup \mathcal{O}(V_1,\beta) \subset \mathcal{O}(\{V_t\},\beta)$$
such that no compact open subset of $\mathcal{O}(\{V_t\},\beta)$ contains y.

Equivalently, the endpoint orbit string y cannot be included in any compact continuation block through the family.

For suitably regular families, this agrees with the preliminary Definition 1.18.

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